

Volatility and Under-Insurance in Economies with Limited Pledgeability: Evidence from a Frost Shock*

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Abstract: How does limited pledgeability shape risk management and access to external finance after aggregate shocks? We show that collateral constraints link two issues that are usually studied separately: firms are under-hedged before shocks despite subsidized insurance, and afterwards credit is least available afterward to the firms with the largest losses. In our dynamic limited-commitment model, insurance uses debt capacity that could otherwise finance production. Insurance and credit compete ex ante, but become complements ex post, because insurance preserves net worth and borrowing capacity. When credit is backed by output and capital at risk, equilibrium price feedback shifts borrowing capacity toward firms whose productive assets survive. We test this mechanism using administrative microdata on bank credit, interfirm payments, and crop insurance around Brazil's July 2021 frost. Coffee trees are productive capital and collateral, so the frost is a shock to cash flow and pledgeable value. Although premiums average 30% below actuarially fair levels, only 20% of coffee farmers were insured. After two years, each percentage point of damage is associated with 1.4% less debt and 0.38% less spending on upstream inputs among uninsured farmers, while insured farmers increase input purchases by 0.24%. Default and interest rates were unchanged. Debt adjusted through forbearance and quantities. An implication of our model is that climate stress tests that interpret low bank losses as resilience mis-measure who bears the risk.

JEL Classification: G32, G22, G21, Q54, Q14, E44.

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1 Introduction

Collateral constraints dynamically link firms' ex ante risk management to their ex post access to external finance. Firms must pay for insurance or other state-contingent contracts before state realization, using the same pledgeable capacity that finances investment. This implies that insurance and credit compete ex ante (Rampini and Viswanathan, 2022). Afterward, insurance and credit become complements. Insurance payouts preserve net worth and pledgeable collateral, sustaining borrowing capacity precisely when unhedged firms face binding credit limits. This dynamic interaction generates a joint allocation failure. Before a shock, financially constrained firms remain systematically under-hedged because allocating scarce borrowing capacity to insurance sacrifices current productive scale. After a shock, credit is least available to the firms that sustained the largest losses, because their collateral has been destroyed. Globally, crop insurance subsidies exceed \$20 billion yet under-insurance remains pervasive (Hazell and Varangis, 2020; Kramer et al., 2022; Sastry et al., 2024), while post-disaster emergency credit routinely fails to reach the most damaged producers. We argue that both issues can arise from a single friction: limited pledgeability.

Our theoretical contribution is to show this mechanism when a disaster destroys the pledgeable asset and the equilibrium output price responds to aggregate losses. In standard models of collateralized borrowing, collateral values are evaluated in ordinary states, when the pledged asset is intact and its market price is determined externally. When an aggregate shock destroys productive capital, however, it simultaneously impairs cash flow and destroys collateral value. Furthermore, when the shock reduces industry output, the contraction in market supply raises the equilibrium output price. We characterize the recursive equilibrium of a dynamic limited-commitment economy with endogenous inverse demand to show that this price increase operates through two distinct channels: a budget channel that raises revenue on surviving output, and a collateral channel that raises the pledgeable value of surviving assets. The loss of a firm's own collateral is distinct from the pecuniary price effect on other firms' borrowing constraints (Dávila and Korinek, 2018). This distinction implies that an aggregate shock widens the cross-sectional dispersion in borrowing capacity between damaged and undamaged firms, rather than simply shifting the distribution.

The model shows that the financial incidence of an aggregate shock depends directly on the contractual composition of collateral. We consider borrowing secured by future harvest revenue through the *Cédula de Produto Rural* (CPR), a registered note in which the farmer promises future delivery of the crop, or its cash value, in exchange for funds today. We compare CPR-secured borrowing with land-secured borrowing. We separate

land value into bare land, which is unaffected by physical destruction, and the standing coffee grove. If credit is backed by future harvests, pledgeable capacity contracts with crop loss but recovers with the higher output price. If credit is backed by bare land, collateral value provides a stable floor unaffected by physical damage. If credit is backed by land appraised inclusive of the perennial grove, the destruction of trees directly reduces pledgeable land value, while the price increase buffers surviving groves. In the calibrated and solved fixed-price benchmark, we show that the optimal state-contingent contract can be implemented through endogenous borrowing limits, similarly to Alvarez and Jermann (2000), but without permanent exclusion from credit markets after default. The solved model generates the event studies that motivate our empirical tests and shows why emergency credit subject to the same collateral constraint fails to expand borrowing capacity for the worst-damaged firms.

We test this mechanism using administrative microdata on bank credit, interfirm payments, and crop insurance around Brazil's July 2021 frost, an extreme weather event that destroyed productive capital across Brazil's main coffee-growing states. Coffee is a perennial crop whose trees take several years to mature. For coffee farmers, coffee trees are a core component of farm capital and pledgeable collateral, so the frost reduced collateral value in addition to current output. We combine transaction-level administrative data from the Central Bank of Brazil: the credit registry (SCR), interfirm payment records, the rural credit registry with its linked public crop insurance program (SICOR), and data from the Ministry of Agriculture (MAPA) on contracts from the main private crop insurers. We link credit, insurance, and firm-to-firm payment microdata for the same farmers, including small farmers who borrow under a personal identifier rather than a firm identifier, which we believe we are the first to do comprehensively in any setting.

This setting is well-suited because it shuts down the three leading alternative explanations for under-insurance and credit contraction. First, insurance access and pricing do not explain low coverage: every farmer can access public revenue insurance, most can choose to buy additional insurance from private providers, and premiums average 30 percent below actuarially fair levels, yet coffee take-up is only 20 percent. Second, the interest-rate and bank-distress channels do not drive the credit contraction: agricultural lending in Brazil is dominated by a state-backed lender operating under administered lending rates, shutting down the risk-spread adjustments and intermediary solvency crises that confound market-rate settings. Third, local demand, weather expectations, and selection into insurance are absorbed by comparing farmers within the same municipality and the same initial insurance status, exploiting granular spatial variation in a frost that forecasts did not resolve until days before it landed, after planting and insurance decisions were sunk.

The model implies the empirical specification and rationalizes four findings. We estimate event studies on credit, input spending, and insurance outcomes, comparing more and less affected farmers within municipality and initial insurance status. First, uninsured farmers experience larger declines in borrowing after the shock than insured farmers: among uninsured coffee farmers, experiencing a 1 percentage point more severe shock is associated with 1.4 percent less debt in the two years after the shock. Second, uninsured farmers reduce input spending more, with each percentage point of shock severity associated with 0.38 percent less expenditure on upstream inputs used to rebuild capital, while affected insured farmers instead increase input purchases by 0.24 percent. Third, we find no increase in default rates, and lenders grant forbearance instead, extending outstanding debt maturity by 0.2 months per percentage point of damage, roughly five months at the average damage of 27 percent. Fourth, among observed policies, insured value per hectare subsequently falls with frost severity for coffee. The corresponding relationship is absent for annual crops such as corn and wheat, which are replanted each season and do not embody lasting grove value. We also use pre-shock guarantee composition to test whether the borrowing response to frost damage varies with the share of debt backed by land rather than harvest revenue. This design maps the model's collateral types into the credit registry without treating guarantee choice as randomly assigned. These findings are inconsistent with models that treat credit and insurance as substitutes. Instead, they confirm that borrowing capacity and insurance interact dynamically through collateral constraints. Unlike models with endogenous exclusion or interest rate spreads, our framework generates under-insurance through collateral limits in a setting where borrowers are not excluded after default and loan rates are largely administered.

The model provides two implications for economic policy. The first implication is that post-disaster credit cannot substitute for insurance when it is intermediated under the same collateral constraint. Following the frost, the Central Bank of Brazil allocated 1.3 billion BRL (approximately 250 million USD) to be disbursed as emergency credit lines to affected coffee farmers. Only around half was lent, and 40 percent of the lending went to the 20 percent of farmers who already held insurance. In the model, an announced credit line does not expand the borrowing capacity of a farmer whose collateral has been destroyed. We consider two methods to relax the constraint. One is public guarantees that remove part of the promised repayment from the farmer's collateral constraint. Another is registries or monitoring that raise the pledgeability of harvest proceeds and land.

The second implication is that the framework changes how to interpret a climate stress test. Supervisory exercises, including those coordinated through the Network for Greening the Financial System, assess climate risk primarily through direct banking-sector losses. In our setting, default and interest rates did not move, consistent with lenders anticipat-

ing state-contingent repayment and granting maturity extensions. Low measured bank exposure did not indicate resilience. The risk stayed with farmers, who cut investment, and with their suppliers, who lost sales. Under the model’s inelastic-demand benchmark, the reduction in aggregate supply passes part of the loss to consumers through the coffee price. A stress test that monitors only financial-sector solvency can miss how the real sector absorbs the shock.

Our model contributes to the theoretical financial literature on the causes and consequences of under-insurance. Froot et al. (1993) establish that hedging matters because investment and financing policies must be coordinated when external finance is costly. Caballero and Krishnamurthy (2003) show that financially constrained firms undervalue insurance when the marginal value of liquidity in production exceeds the market price. Dávila and Korinek (2018) show how financial frictions generate distributive and collateral externalities. Martins-Da-Rocha et al. (2022) emphasize the role of limited pledgeability and strategic default. Unlike models where amplification operates through the interest rate, our mechanism operates with a fixed funding rate through changes in the output price and its effect on collateral. Sastry et al. (2024) document a comparable protection gap in US property insurance and provide evidence consistent with the Rampini and Viswanathan (2022) mechanism, but focus on household insurance demand. In contrast, we study firms’ borrowing and investment responses to a realized shock to productive capital and collateral.

More broadly, our paper adds to a literature documenting that financial frictions amplify the real effects of shocks through lower investment, distorted insurance decisions, and endogenous price dynamics. Alfaro et al. (2024) show that financial frictions more than double the impact of aggregate uncertainty shocks on investment and financial variables, whereas our mechanism turns on the interaction of capital destruction, collateral constraints, and price feedback in general equilibrium. Other papers, such as Arellano et al. (2019), Ottonello and Winberry (2020), and Giroud and Mueller (2017) show how financial frictions propagate shocks through investment, credit, and labor channels, but do not study the role of hedging. Our model and regressions show that under-insurance can amplify the real effects of a capital shock through both credit and investment.

In addition to this paper’s contribution at the intersection of model and empirics, this paper contributes to two largely empirical strands of the finance literature. The first is an empirical literature on the interaction between credit and insurance. One strand focuses on the role of ex ante financial constraints that can be alleviated by financing the insurance premiums (Cole and Xiong, 2017; Casaburi and Willis, 2018). In our setting, subsidized lending is linked to subsidized insurance premiums, which McIntosh et al. (2013) show increases willingness to pay for insurance but Annan (2022) show can exacerbate moral

hazard. Yet insurance take-up is still low, so we believe that financing premiums is not a sufficient explanation. Lane (2024) follow a long literature that studies whether credit can substitute for insurance. We show that credit and insurance compete for debt capacity ex ante but are complements ex post, and we show how this arises from collateral constraints that bind ex post rather than ex ante. This analysis draws on a comprehensive dataset linking credit and insurance from both formal (firm) and informal (individual) contracts with banks and insurers at the transaction level.

The second is a long empirical literature in corporate finance, including a burgeoning literature in climate finance, on the impact of weather shocks on farmers through financial channels. Bergman et al. (2020) estimate the elasticities of land prices and revenue to yields using a weather shock instrument, in the setting of the farm debt crisis in the 1980s in Iowa before the introduction of crop insurance. Brown et al. (2021) estimate how firms respond to severe weather shocks using credit lines, and find that heavier snow at a firm's headquarters decreases cash flow and increases short-term lending with no impact on investment nor capital stock. Cortés and Strahan (2017) show that there is reallocation of lending across locations in response to a natural disaster shock as banks' balance sheets deteriorate. Benincasa et al. (2024) document how firms finance losses from extreme weather events, and Giroud et al. (2012) use exogenous weather variation to show how leverage constrains investment. Weagley (2019) shows that intermediary capacity determines the price of weather risk transfer, which is the supply-side counterpart to the demand-side constraint that we study. Deschênes and Greenstone (2007) estimate the effect of weather on agricultural output without the financial channel that is central here. In our setting, there is a dominant nationwide agricultural sector lender backstopped by the government, so lending rates did not increase after the shock, allowing us a clean test of the insurance mechanism. In general, our setting and mechanism differ from each of these papers: the interaction of the physical shock, causing a reduction in the capital stock, with insurance generates our financial and hedging results.

Netto (1959) was the first to describe the coffee cycle formally, using a theoretical model and historical data to show how expansion and crisis alternate because farmers respond to past prices and coffee plants take time to mature. Our model provides a different mechanism with forward-looking farmers: a frost raises coffee prices at the same time that it destroys productive capital, so the price increase relaxes borrowing constraints most for farmers whose capital and price-sensitive collateral survive, while affected constrained farmers may rebuild slowly. The model event study illustrates how this difference can make a one-time shock persistent.

The remainder of the paper is organized as follows. Section 2 presents the model and the model-generated event studies. Section 3 describes the institutional setting in Brazil,

the data, and the timeline of the frost shock. Section 4 introduces the empirical methodology and presents the regression results. Section 5 considers policy implications for emergency credit, collateral pledgeability, and climate stress testing. Section 6 concludes.

2 Model

To interpret the empirical patterns, we develop a dynamic model with capital shocks, borrowing constraints, and two forms of collateral. We build upon Rampini and Viswanathan (2013, 2022), but consider a production economy that experiences both aggregate and idiosyncratic capital-stock shocks similar to Caballero and Krishnamurthy (2003) and Lorenzoni (2008). A CPR pledges future harvest revenue, while a land-secured loan pledges the value of bare land and, depending on the appraisal, the standing coffee grove. We use fixed-price experiments to isolate how these collateral structures propagate a common shock and then characterize the equilibrium price channel through inverse demand.

The distinction between these forms of collateral matters after a frost. The frost destroys productive capital and reduces the future harvest, but does not destroy bare land. At the same time, the aggregate reduction in coffee supply raises the coffee price. The price increase raises current revenue and the value of the harvest and grove that survive the shock. The model therefore features both a budget channel, through the effect of price on farmers' internal funds, and a collateral channel, through the effect of price on pledgeable value. These channels can offset part of the aggregate loss while increasing differences across farmers with different physical damage and collateral.

The recursive environment nests three useful cases. Setting $\lambda = \chi = 0$, allowing direct recovery of productive capital, and taking the price process as given produces the capital-constraint benchmark derived in Appendix C.4. Setting $\lambda_K = \chi = 0$ gives the CPR-only model with an output-based borrowing constraint. Its solved fixed-price household problem produces the paired event studies below. Setting $\lambda_K = 0$ and allowing both $\lambda > 0$ and $\chi > 0$ adds land collateral and separates bare land from the standing grove. We feature the CPR-and-land constraint in the exposition because it describes the collateral used by farmers.¹

¹All reported quantitative regime comparisons hold the price sequence fixed so that differences reflect collateral structure rather than a simultaneous change in aggregate prices.

2.1 Environment and Timing

There is a continuum of risk-averse farmers indexed by $i \in [0, 1]$. Farmer i has permanent productivity type $a \in \{L, H\}$, with productivity A_a , and preferences

$$U_i = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_{it}) \right], \quad u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \quad (1)$$

where $\beta \in (0, 1)$ and $\sigma > 0$.

At the beginning of a period, a farmer has net worth W , aggregate state S , and individual state z . The farmer chooses consumption c , installed capital k , and a schedule of one-period state-contingent net repayments $b(S', z')$. The aggregate state determines the incidence of frost damage, while the individual state determines the realization of the capital-survival rate $\theta(z', S')$. We distinguish these states because aggregate prices are publicly observed, whereas plot-level damage is established through agronomic verification. The unrestricted repayment schedule below is the optimal-contract benchmark. We subsequently impose the information structure of observed crop insurance. Under this, ordinary credit can condition on the public aggregate state, but policies under the Rural Insurance Premium Subsidy Program (PSR) condition on verified plot-level damage.

The joint state follows a Markov process with transition probability $\pi(S', z' \mid S, z)$. Aggregate frost incidence determines the probability that an individual farmer receives the capital shock, so the model contains both aggregate and idiosyncratic risk.

Capital produces coffee after the next shock is realized:

$$y_a(S', z') = A_a \theta(z', S')^\alpha k^\alpha, \quad \alpha \in (0, 1), \quad (2)$$

and $\theta(z', S')k$ units of productive capital survive into the next period. Thus k is the pre-shock installed stock. The same physical frost realization lowers both the surviving grove and the harvest produced by that grove. In the quarterly computation, the normal-state value $\theta_n = 1 - \delta$ incorporates depreciation, while $\theta_f < \theta_n$ is the survival rate under frost.

Financial intermediaries are risk neutral, have deep pockets, and discount at the gross rate R , with $\beta R < 1$. Rural lending rates in our setting are largely administered, which motivates an exogenous R . In the optimal-contract benchmark, intermediaries price a net-repayment schedule over aggregate and individual states, while limited commitment restricts what they can recover following default. This spanning assumption isolates the collateral mechanism. The observed-contract formulation below restricts ordinary debt to public states and reserves plot-level contingency for insurer-verified PSR payments. Farmers are not permanently excluded from future credit markets after default.

2.2 Limited Commitment, Debt, and Insurance

Let $d \geq 0$ denote non-state-contingent debt and let $h(S', z') \geq 0$ denote a state-contingent insurance payment. Define net repayment in state (S', z') by

$$b(S', z') \equiv d - h(S', z'). \quad (3)$$

Positive b is net debt repayment, while reducing b in a bad state provides insurance. Debt and insurance therefore determine repayment jointly in every state.

We write the solved benchmarks and the recursive equilibrium environment using one enforcement constraint. Let $\lambda_K \in [0, 1]$ be the recoverable share of surviving productive capital. This is the capital-based constraint in the solved benchmark. A CPR or harvest pledge makes fraction $\lambda \in [0, 1]$ of next-period harvest revenue pledgeable, while fraction $\chi \in [0, 1]$ of appraised land value can support repayment. The state-by-state borrowing constraint is

$$b(S', z') \leq \underbrace{\lambda_K \theta(z', S') k}_{\text{productive capital}} + \underbrace{\lambda p(S') A_a \theta(z', S')^\alpha k^\alpha}_{\text{CPR (harvest) pledge}} + \underbrace{\chi q(S', z')}_{\text{land pledge}}, \quad (4)$$

$$q(S', z') = \bar{L} + \psi G(S', z'), \quad G(S', z') = \kappa_G p(S') \theta(z', S') k. \quad (5)$$

Here $\bar{L} \geq 0$ is the value of bare land, which is not damaged by the frost. The term G is the appraised value of the surviving grove. The parameter $\psi \in [0, 1]$ is the share of grove value included in the appraisal, and κ_G converts surviving coffee capital into appraised value. We interpret χ as a reduced-form recovery rate that combines title quality, foreclosure, and enforcement. The current model does not track the physical transfer and subsequent ownership of land after default. We impose

$$\lambda_K + \chi \psi \kappa_G p(S') \leq 1 \quad \text{for every } S', \quad (6)$$

so that the combined claims against productive capital and grove value do not exceed the replacement cost of surviving capital. We include land in the collateral constraint, but not in the flow budget, because grove capital already produces output and survives into the next period. The variable θk already includes normal depreciation in the recursive model. Regime I has $\chi = 0$ and uses only CPR collateral. Regime II-a has $\chi > 0$ and $\psi = 0$, so land provides a fixed collateral cushion. Regime II-b has $\chi > 0$ and $\psi > 0$, so land value depends on both grove damage and the coffee price. To isolate the collateral effect of grove valuation from the overall scale of pledged assets, we equate normal state land capacity across Regimes II-a and II-b. Let $G_n = \kappa_G p_n \theta_n \bar{k}$ denote normal grove value for a reference

capital stock and impose

$$\bar{L}^{\text{IIa}} = \bar{L}^{\text{IIb}} + \psi G_n. \quad (7)$$

Without this adjustment, the regime comparison would also shift the initial level of collateral. We evaluate $\bar{k}$ as a normalization uniform in wealth over the non-negative constrained policy range of Regime II-b. This benchmark fixes total normal land capacity without creating circular dependence on simulated sample distributions.

Given this enforcement limit, any state-contingent net repayment can be written as debt minus insurance.

Proposition 1 (Debt–Insurance Representation). Any bounded net-repayment schedule satisfying the enforcement limit can be implemented with one-period non-state-contingent debt d and nonnegative state-contingent insurance $h(S', z')$ satisfying

$$d \leq \lambda_K \theta(z', S') k + \lambda p(S') A_a \theta(z', S')^\alpha k^\alpha + \chi q(S', z') + h(S', z') \quad \text{for every } (S', z'). \quad (8)$$

Conversely, any such debt and insurance pair induces a net-repayment schedule satisfying the enforcement limit.

Proof. See Appendix C. The construction uses $b = d - h$ and the recovery value in (4). $\square$

Under Proposition 1, we interpret any net-repayment schedule as a debt contract plus an insurance contract. A more state-contingent schedule preserves net worth after a bad shock but reduces the funds available before it. Farmers with low net worth choose schedules that leave them partly unhedged even when state-contingent claims are actuarially fair, because each unit of insurance reduces the debt available to finance capital.

2.2.1 Observed PSR contracts and participation

We map the benchmark into the observed insurance institution by distinguishing public aggregate states from insurer-verified plot damage. Let $v \in \mathcal{Q} = \{0, 0.65, 0.70\}$ denote no policy or one of the two coverage levels that account for nearly all insured coffee policies, with $H(0, z') = \Pi(0) = 0$. Ordinary debt may condition on the public aggregate state, $d(S')$, but not directly on the agronomist's plot-level productivity assessment. A PSR policy supplies that missing contingency through a product-specific indemnity $H(v, z')$ and requires an upfront farmer premium $\Pi(v)$. For a yield policy, the basic payoff takes the form

$$H_i^Y(v, z') = \bar{p}_i L_i [\bar{y}_i - y_i(S', z')]^+, \quad (9)$$

where $\bar{p}_i$ is the price fixed in the policy, L_i is insured area, and $\bar{y}_i$ is expected productivity. A cost policy uses the same verified productivity shortfall to determine reimbursement

against its insured production-cost limit. Thus the frost jointly reduces grove capital and future yield, while the contract pays only according to its stated yield or cost-loss schedule.

Let $\rho_H \in [0, 1]$ denote the share of a verified indemnity that can be pledged or assigned to the lender, and write $\mathcal{B}_a(S', z'; k)$ for the collateral value on the right-hand side of (4). Conditional on policy v , the institutional problem replaces unrestricted plot-contingent repayment with

$$c = W + \mathbb{E} \left[\frac{d(S')}{R} \mid S, z \right] - k - \Pi(v), \quad (10)$$

$$W' = p(S')A_a\theta(z', S')^\alpha k^\alpha + \theta(z', S')k + H(v, z') - d(S'), \quad (11)$$

$$d(S') \leq \mathcal{B}_a(S', z'; k) + \rho_H H(v, z') \quad \text{for every } z'. \quad (12)$$

The farmer chooses among the three policies by comparing the conditional value functions. This restriction gives PSR an economic role without requiring all debt to be non-contingent: debt responds to public aggregate conditions, while insurance responds to verified farm damage. When $\rho_H < 1$, the premium absorbs current financing capacity before the indemnity becomes fully pledgeable. Collateral constraints can then reduce participation even when the subsidized contract has positive expected value.

The quantitative collateral-regime exercise below uses the unrestricted benchmark to isolate the propagation mechanism. We match the observed pre-shock insurance share through the mass assigned to the higher-coverage group and hold group membership, farmer draws, and shock assignments fixed across regimes. The resulting horse race is a fixed-composition comparison of collateral structures. The observed-contract block provides the institutional interpretation of insurance participation, while the reported quantitative exercise identifies how collateral propagates the frost conditional on the pre-shock composition.

2.3 Recursive Equilibrium and Price Determination

Farmers take the perceived laws of motion for the distribution and price as given. We suppress the distribution in the notation: $V_a(W, S, z)$ denotes $V_a(W, S, z; \mu)$, and $p(S')$

denotes the perceived price $p(S', \mu')$. The farmer solves

$$V_a(W, S, z) = \max_{k, \{b(S', z')\}} \left\{ u(c) + \beta \mathbb{E} [V_a(W', S', z') \mid S, z] \right\} \quad (13)$$

$$\text{s.t.} \quad c = W + \mathbb{E} \left[\frac{b(S', z')}{R} \mid S, z \right] - k, \quad (14)$$

$$W' = p(S') A_a \theta(z', S')^\alpha k^\alpha + \theta(z', S') k - b(S', z'), \quad (15)$$

$$b(S', z') \leq \lambda_k \theta(z', S') k + \lambda p(S') A_a \theta(z', S')^\alpha k^\alpha + \chi q(S', z') \quad \forall (S', z').$$

Net worth is the liquid state variable relevant for current choices. The state-contingent repayment schedule determines endogenous hedge coverage in the benchmark. Binary PSR participation is instead the discrete contract choice defined by equations (9)–(12).

Let μ_t be the distribution of farmer type, wealth, and individual state before choices are made. Given the capital policy k_{it} , next period's output and price are

$$Y_{t+1} = \int_0^1 A_{a_i} \theta(z_{i,t+1}, S_{t+1})^\alpha k_{it}^\alpha di, \quad (16)$$

$$p_{t+1} = c_p Y_{t+1}^{-1/\gamma}, \quad \gamma > 0. \quad (17)$$

The parameter γ is the price elasticity of demand in absolute value. In equilibrium, farmer policies generate the distribution of installed capital, and the resulting output and price satisfy (16)–(17).

A recursive competitive equilibrium consists of farmer value and policy functions, a law of motion for μ , and a price function such that: (i) farmer policies solve (13) subject to the budget and collateral constraints; (ii) intermediaries break even by pricing the repayment schedule at $\mathbb{E}[b(S', z')]/R$; (iii) the policy functions and shock process generate the next-period distribution; and (iv) aggregate output and the coffee price satisfy (16)–(17). In particular, equilibrium requires

$$\mu' = \Phi(\mu, S, S'; k_a, b_a), \quad p(S', \mu') = c_p Y(S', \mu')^{-1/\gamma}, \quad (18)$$

where Φ is generated by the policy functions and the joint shock process. The equilibrium price is therefore an outcome of farmers' investment choices and the distribution of frost damage.

A widespread frost reduces output and raises the coffee price. The budget channel operates because the higher price raises revenue and next-period net worth for farmers with surviving production. The collateral channel operates because the same price enters both the CPR pledge and the grove component of land value. Holding capital fixed, the

response of the borrowing limit to the price is

$$\frac{\partial \mathcal{B}_i(S', z'; k_i)}{\partial p(S')} = \lambda A_{a_i} \theta_i(S', z')^\alpha k_i^\alpha + \chi_i \psi_i \kappa_G \theta_i(S', z') k_i. \quad (19)$$

The price increase therefore relaxes borrowing constraints most for farmers with more surviving harvest and grove capital.

Let $\xi_i(S', z')$ be the multiplier on farmer i 's collateral constraint. We summarize collateral-price exposure, holding policies and the distribution fixed, by

$$\Lambda^p(S') = \int \sum_{z'} \xi_i(S', z') [\lambda A_{a_i} \theta_i(S', z')^\alpha k_i^\alpha + \chi_i \psi_i \kappa_G \theta_i(S', z') k_i] d\mu(i). \quad (20)$$

Only farmers with binding constraints contribute to $\Lambda^p(S')$. It separates the direct effect of price on collateral from the effect of price on current revenue. The full equilibrium can therefore generate a partial aggregate offset at the same time that it widens the difference between affected farmers and farmers whose collateral survives.

2.4 Benchmark Parameterization and Numerical Solution

We solve the fixed-price household problems by iterating on the marginal value of wealth. At each wealth point, the algorithm evaluates candidate binding sets, retains only choices that satisfy feasibility and complementary slackness in every continuation state, and interpolates the resulting policies over net worth. We then simulate the same farmers, shock assignments, and aggregate price sequence under each collateral regime and apply the empirical event-study transformation to the simulated panel. Holding these objects fixed makes the regime comparison a direct measure of collateral propagation.

The household solver is the endogenous grid method of ?, extended to occasionally binding collateral constraints as in ?. For the general equilibrium model, we add the coffee price as an aggregate state variable and close the model with the simulation-based forecasting rule of ?. As in ?, where collateral-constrained producers forecast an equilibrium price rather than aggregate capital, farmers forecast next period's log coffee price with a rule that is linear in the current log price and whose coefficients depend on the current and next frost states. The continuation marginal value of wealth is evaluated at the forecast price in each continuation state. We simulate the cross-section, clear the market each period through (17), re-estimate the rule, and iterate with damping until its coefficients converge. The simulation reuses the same aggregate and idiosyncratic draws in every iteration (?), so changes in the estimated rule reflect changes in policies rather than sampling noise. A high R^2 is insufficient to show the accuracy of the forecasting rule, so we report

the statistic of $\hat{p}_t$, which iterates the rule forward on the realized frost sequence without resetting it to the simulated price. In the benchmark, the maximum absolute error in the log price over 2,500 simulated periods is 0.14 percent (mean 0.02 percent). With perfectly elastic demand ($\gamma \rightarrow \infty$), the solver reproduces the fixed-price solution to within 10^{-6} . This equilibrium exercise uses a two-state calibration with $\lambda = 0.7$, $\beta = 0.9$, $R = 1.05$, $\theta_f = 0.5$, and demand elasticity $\gamma = 0.4$. The collateral-regime comparison below instead holds the price path fixed across regimes, so that differences across regimes reflect collateral structure alone.

Table 1 reports the calibrated inputs used in the four-state fixed-price CPR exercise. The annual discount factor, funding rate, and depreciation rate are converted to the quarterly frequency. Table 2 reports the additional inputs used to compare CPR collateral with fixed and grove-inclusive land collateral. The separate capital-constraint benchmark in Appendix C.4 uses the parameter values in Table 6.

Table 1: Inputs for the Fixed-Price CPR Exercise

Parameter	Value	Interpretation
α	0.30	Capital elasticity
λ	0.40	Pledgeable share of harvest revenue
β^{ann}	0.85	Annual discount factor
R^{ann}	1.10	Annual gross funding rate
σ	2.00	CRRA coefficient
δ^{ann}	0.05	Annual normal depreciation
θ_f	0.30	Capital survival under frost

At the quarterly frequency, we use

$$\beta = (0.85)^{1/4}, \quad R = (1.10)^{1/4}, \quad \delta = 1 - (0.95)^{1/4}. \quad (21)$$

The frost survival rate θ_f is event-specific and is not annualized.

Table 2: Benchmark Inputs for Land Collateral

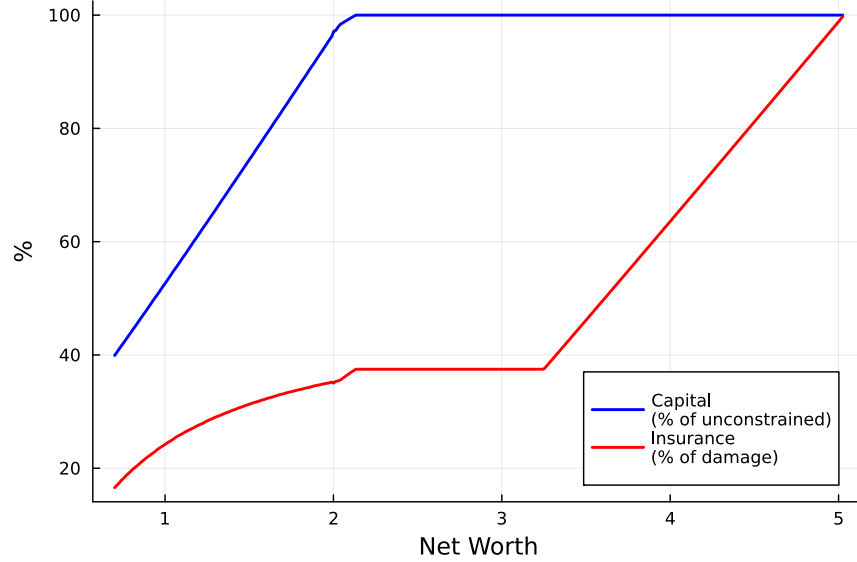
Parameter / Identified Object	Value	Source / Identification
Bare-land share	83.64%	INCRA 2022 Minas Gerais coffee report (VTN / VTI)
Residual grove share	16.36%	Upper-bound proxy for trees and improvements
Prudential LTV ceiling (χ)	0.60	Central Bank of Brazil Vote 127/2019
Grove pledge recognition (ψ)	1.00	Full grove pledgeability endpoint (Regime II-b)
Reference capital stock ($\bar{k}$)	0.555	Normalization uniform in wealth on constrained II-b
Pledged normal land capacity	0.334	Equated to CPR capacity at reference capital stock
Identified bare-land floor ($\chi\bar{L}$)	0.279	Product in borrowing constraint
Identified grove slope ($\chi\psi\kappa_G$)	0.100	Product in borrowing constraint

The household borrowing constraint identifies land collateral through two effective products: the bare-land floor $\chi\bar{L}$ and the grove slope $\chi\psi\kappa_G$. We discipline these objects using public administrative benchmarks. Table 2 reports the inputs. We split appraised property values using the 2022 INCRA land-market report for Minas Gerais coffee municipalities. Bare land represents 83.64% of total appraised property value (VTN of R\$71,310 per hectare versus total property value VTI of R\$85,261 per hectare). The 16.36% residual provides an upper bound for the grove share. To map effective products back to asset levels, we use Central Bank of Brazil Vote 127/2019, which limits first-lien rural real estate debt to 60% of appraised collateral ($\chi = 0.60$). Finally, we normalize normal pledged land capacity to equal the CPR limit evaluated at $\bar{k} = 0.555$, calculated uniformly over wealth on the constrained policy space. Holding these effective products constant, shifts in χ leave solved policies unchanged.

To connect these collateral technologies to the credit registry, we allow farmer type to include both productivity and the composition of pledged collateral. Harvest-backed, land-backed, and mixed farmers differ in their effective CPR and land capacity. The observed land-guarantee share is therefore informative about relative effective collateral capacity, rather than about χ , $\bar{L}$, ψ , or κ_G separately. In the simulated cross-section, we assign one collateral type to each farmer using the pre-shock empirical type shares and apply the same frost-by-land-share event-study estimator that we use in the data. We report this model-to-data comparison separately from the dynamic horse race below, which instead applies each collateral regime to the same farmers in order to isolate the collateral mechanism.

Figure 1 shows that our model replicates the Rampini and Viswanathan (2013, 2022) result on net worth and hedging. The figure plots capital as a percentage of its unconstrained level, and insurance as a percentage of damage, against net worth. Farmers restore scale before they hedge. At intermediate net worth, they operate at the unconstrained capital level but leave most of the damage uninsured.

Figure 1: Model-Implied Relationship Between State-Contingent Coverage and Net Worth



Notes: This figure shows the model-implied relationship between net worth and state-contingent coverage. Under incomplete markets, a binding pledgeability constraint reduces optimal hedging at low net worth relative to the complete-markets benchmark. Coverage is defined from the net-repayment schedule and is distinct from binary PSR participation.

2.5 Model Event Study

The paired fixed-price CPR model event study uses four aggregate nodes that combine low or high frost incidence with a low or high coffee price.

Table 3: Aggregate States in the Model Event Study

State	Coffee Price	Probability of Frost Damage
s_1	1.0	s_n
s_2	1.5	s_n
s_3	1.0	s_f
s_4	1.5	s_f

Assume $s_f > s_n$, where f denotes frost and n denotes normal. For the fixed-price CPR benchmark, we set $s_n = 0$ and $s_f = 0.5$. These values are distinct from the 20% and 50% incidences in the separate two-state capital-collateral benchmark in Appendix C.4. Their unconditional distribution is $P_S = (0.1, 0.1, 0.6, 0.2)$. In the fixed-price exercise, the aggregate state is i.i.d., so each row of the transition matrix equals P_S . This four-node exercise separates farmers by initial model coverage.

We use the numerical solution of the fixed-price CPR household problem to construct

the paired event study that motivates our empirical specification. The benchmark sets $\lambda_K = \chi = 0$ and uses the four-node coffee-price process. We feed the model the following sequence of aggregate states:

$$S = \left[1, \underbrace{4}_{\text{frost and high price}}, \underbrace{2, 2}_{\text{high price}} \right]. \quad (22)$$

The first period is a normal state. In the second period, the economy enters the frost state, in which the probability of capital damage and the coffee price are both high. In the last two periods, the probability of damage returns to its normal level while the coffee price remains high.

We measure insurance by how much of the frost-induced gap in resources the state-contingent repayment schedule undoes. Writing $Y(S', z') = p(S')A\theta(z', S')^\alpha k^\alpha + \theta(z', S')k$ for gross resources before repayment, coverage in an aggregate state with damage risk is

$$\text{Cov}(S') = \frac{b(S', z_n) - b(S', z_f)}{Y(S', z_n) - Y(S', z_f)} \quad (23)$$

which equals one when next-period net worth is equalized across the idiosyncratic shock and zero when repayment is not state-contingent. We report the P_S -weighted average over the states in which damage can occur.

Whenever the collateral constraint binds in both nodes of a damage-bearing state, $b = B$. With $\lambda_K = 0$ and no damage-sensitive land component, the harvest pledge is the only damage-sensitive term in the limit, so (23) collapses to

$$\text{Cov}(S') = \frac{\lambda}{1 + \frac{(\theta_n - \theta_f)k^{1-\alpha}}{p(S')A(\theta_n^\alpha - \theta_f^\alpha)}} < \lambda. \quad (24)$$

Coverage at or above λ rules out both CPR constraints binding in every state with damage. The converse is not true. A constraint can slacken while coverage remains below λ , and coverage can exceed λ as it approaches full insurance. We use this comparison to diagnose which constraints bind. Conditional on fixed k and both constraints binding, a shock-insensitive land floor $\chi\bar{L}$ cancels from the repayment difference in (23). After the household problem is re-solved, fixed land collateral can change coverage by relaxing the feasible repayment schedule, changing which constraints bind, and changing capital. We compare the solved regimes directly. We draw 2,000 farmers from a single uniform distribution of initial net worth on $[0, \bar{W}]$ and define the top 20% of the model-coverage distribution as the higher-coverage group. This composition matches the 20% pre-shock

coffee insurance share observed in the registry without using policy participation to identify a preference or enforcement parameter. We assign half of the farmers in each group to receive the individual capital shock. We then track the same farmers, group membership, and shock assignments through all three collateral regimes. Conditional on initial coverage group, the difference between shocked and non-shocked farmers is generated by the capital shock. Differences across regimes isolate collateral structure. We focus on low and high endogenous hedge coverage. By comparison, binary PSR take-up is governed by the separate contract problem above.

Let T denote the simulated frost date, let $\mathcal{T} = \{-1, 0, 1, 2\}$, and define

$$D_{it}^{\tau} = \mathbb{1}\{t - T = \tau\} \text{Shock}_i. \quad (25)$$

We estimate the following event study on the simulated panel:

$$y_{it} = \sum_{\tau \in \mathcal{T} \setminus \{-1\}} \beta_{\tau}^I D_{it}^{\tau} \text{Ins}_i + \sum_{\tau \in \mathcal{T} \setminus \{-1\}} \beta_{\tau}^{NI} D_{it}^{\tau} (1 - \text{Ins}_i) + \alpha_i + \alpha_{gt} + \varepsilon_{it}, \quad (26)$$

where Ins_i denotes the higher-coverage group and g denotes initial coverage group. We normalize the period before the shock to zero and plot cumulative responses.

2.6 Dynamic Horse Race Across Collateral Regimes

We distinguish between installed capital choice k_t and net investment $k_t - \theta_t k_{t-1}$. Capital choice measures scale: the target stock operated in the subsequent period. Net investment measures replenishment outlay: the gross resource flow required to rebuild damaged stock and cover quarterly depreciation. At the impact of the frost ($\tau = 0$), physical survival drops to $\theta_f = 0.30$, destroying 70% of productive trees on shocked farms. For farmers in the higher-coverage group, installed capital choice dips only trivially (-4.35% under CPR and under -0.05% with land collateral). Because their standing trees were destroyed, preserving scale requires an immediate surge in replenishment. Net investment jumps by $+0.582$ to $+0.661$ model units because state-contingent transfers preserve net worth and borrowing capacity for replanting. For farmers in the lower-coverage group, binding collateral constraints prevent full restoration. Capital choice collapses by 55.74% under CPR-only and by nearly 32% under land collateral. Even though higher coffee prices raise the marginal value of capital, lower-coverage farmers commit only $+0.149$ to $+0.300$ units to net investment, rebuilding at less than half the pace of the higher-coverage group.

Table 4: Dynamic Horse Race: Model Impact Responses Across Collateral Regimes

Outcome	Initial Coverage Group	Regime I (CPR)	Regime II-a (Fixed)	Regime II-b (Grove)	II-a minus I	II-b minus II-a
Debt response (% pre-shock)	Higher coverage	−1.29%	+0.35%	+0.24%	+1.65 pp	−0.11 pp
	Lower coverage	−11.75%	−4.72%	−7.17%	+7.03 pp	−2.45 pp
Capital choice (% pre-shock)	Higher coverage	−4.35%	−0.04%	−0.01%	+4.31 pp	+0.04 pp
	Lower coverage	−55.74%	−31.71%	−31.61%	+24.04 pp	+0.10 pp
Net investment ($k_t - \theta_t k_{t-1}$)	Higher coverage	+0.582	+0.659	+0.661	+0.077	+0.001
	Lower coverage	+0.149	+0.290	+0.300	+0.141	+0.010
Coverage response (pp)	Higher coverage	−2.02 pp	−0.02 pp	−0.00 pp	+2.00 pp	+0.02 pp
	Lower coverage	−19.75 pp	−15.84 pp	−15.55 pp	+3.92 pp	+0.29 pp

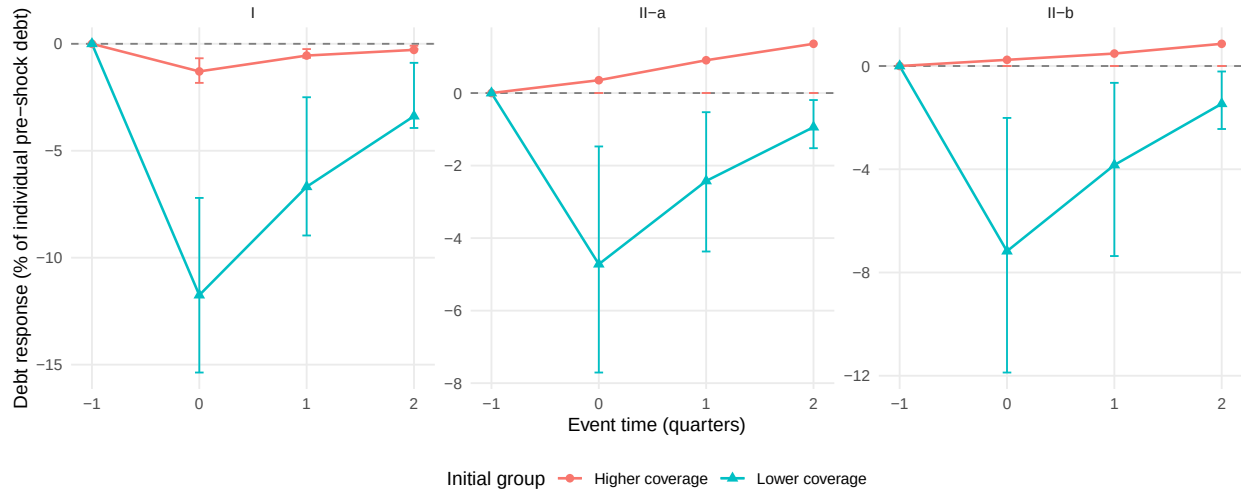
Notes: Event-time $\tau = 0$ regression coefficients from equation (26) estimated on the simulated panel under identical farmer draws and shock assignments. Units: debt and capital choices are expressed as percentages of each farmer’s own pre-shock level; net investment is $k_t - \theta_t k_{t-1}$ in model units; coverage response is in percentage points. Regime I uses CPR collateral only ($\chi = 0$). Regime II-a adds a fixed land floor ($\chi > 0, \psi = 0$). Regime II-b recognizes grove value in land appraisal ($\chi > 0, \psi = 1$).

Table 4 presents the dynamic horse race across the three collateral regimes. Comparing Regimes I, II-a, and II-b delivers two primary takeaways. First, land collateral acts as a vital borrowing cushion. Under CPR-only (Regime I), debt capacity depends strictly on expected harvest revenue. When frost strikes, future output falls, contracting debt among lower coverage farmers by 11.75%. Under fixed land collateral (Regime II-a), that debt contraction shrinks to 4.72%, cutting the borrowing decline by 7.03 percentage points and attenuating the collapse in capital choice by 24.04 percentage points.

Second, fixed bare land protects borrowing capacity more effectively than grove-inclusive land. Comparing Regime II-a with Regime II-b, lower coverage borrowing contracts by an additional 2.45 percentage points under grove valuation (−7.17% versus −4.72%). The reason is structural: bare land does not freeze. Standing coffee trees do. When lenders appraise grove improvements alongside bare land, the frost destroys not only harvest revenue but also the collateral value of the property itself. This dual impairment tightens the enforcement limit at the moment when liquidity needs peak. Once coffee prices rebound, grove-inclusive land creates slightly stronger incentives for forward capital replacement, raising net investment marginally from +0.290 to +0.300 units.

Figure 2 plots the dynamic debt responses across event time for all three regimes. Farmers in the lower coverage group reduce debt persistently under CPR-only. Land collateral mitigates this credit contraction, with fixed land providing the strongest buffer.

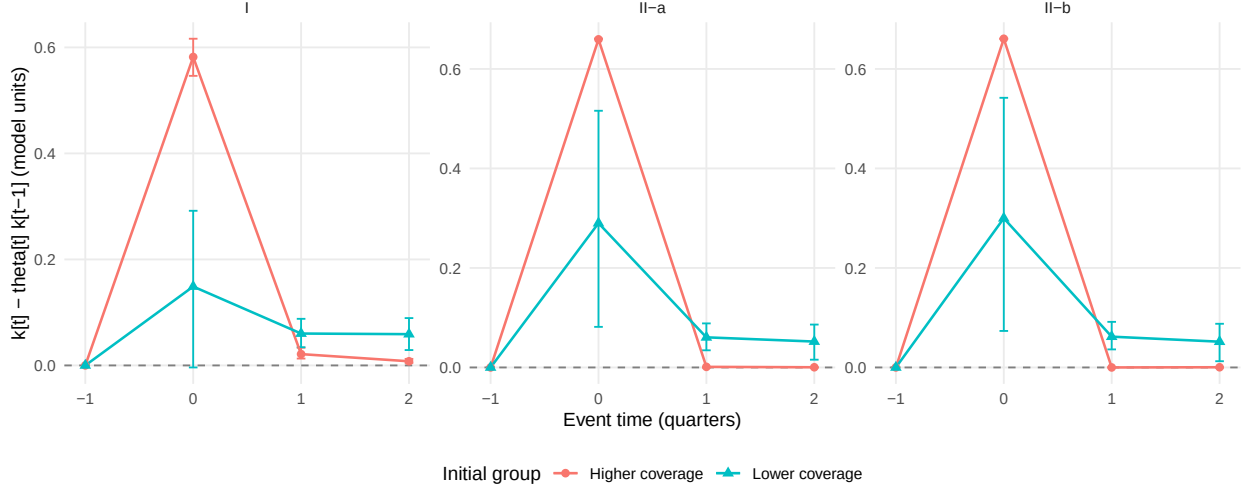
Figure 2: Model-Implied Debt Responses Across Collateral Regimes



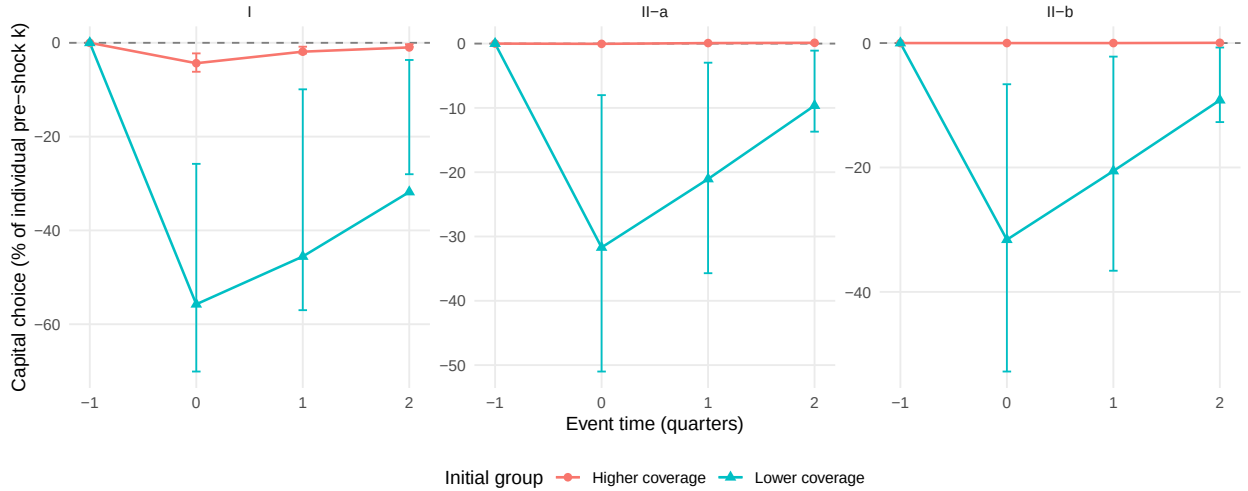
Notes: Model-implied cumulative effects on borrowing across event time, estimated from the simulated farmer panel under aggregate sequence (22). Points are randomized event-study coefficients from equation (26). Error bars show the interquartile range of paired damaged-minus-undamaged effects at identical initial wealth. Farmers are grouped by their initial model coverage in the CPR-only environment. Within each group, half of the farmers receive the capital shock. Regime I uses CPR collateral only ($\chi = 0$). Regime II-a adds a fixed land floor ($\chi > 0, \psi = 0$). Regime II-b recognizes grove value in land appraisal ($\chi > 0, \psi = 1$).

Figure 3 reports the responses of net investment and capital choice. Higher coverage farmers maintain their capital stock by increasing net replenishment immediately after the frost. Lower coverage farmers face binding credit limits that restrict tree replacement, resulting in a persistent capital contraction.

Figure 3: Model-Implied Replenishment and Capital Responses Across Collateral Regimes



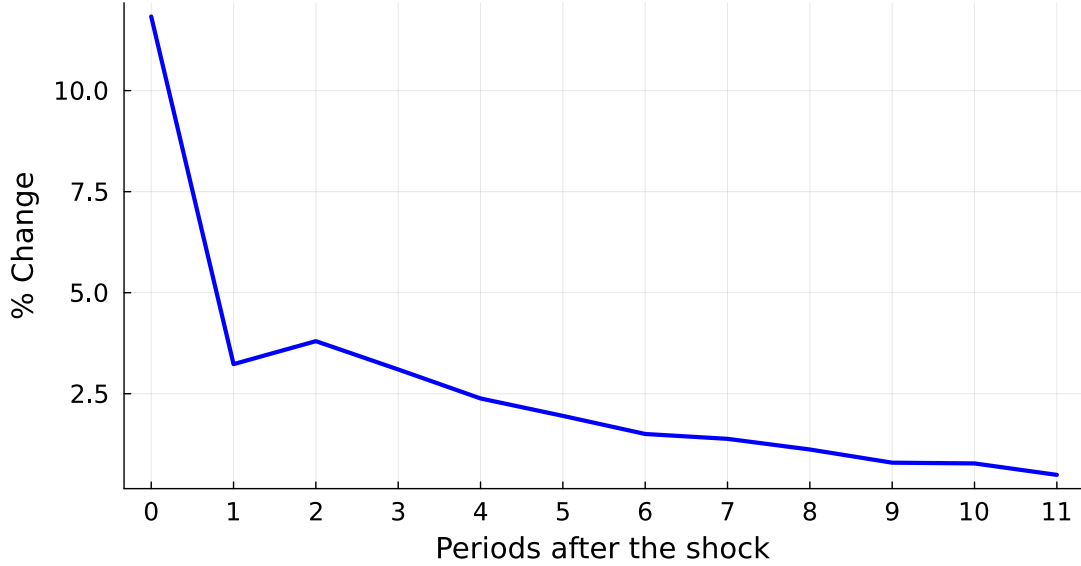
(a) Net Investment ($k_t - \theta_t k_{t-1}$)



(b) Capital Choice (k_t)

Notes: Event-study responses across event time estimated from the simulated panel under aggregate sequence (22). Points are randomized regression coefficients. Error bars report the interquartile range of paired counterfactual differences at identical initial net worth. Panel (a) shows net replenishment $k_t - \theta_t k_{t-1}$ in model units. Panel (b) shows the chosen installed capital stock k_t as a percentage of individual pre-shock capital.

Figure 4: Equilibrium Coffee-Price Response to a Frost
IRF Price with respect to the aggregate shock



Notes: This figure shows the average percentage change in the equilibrium coffee price h periods after the economy enters the frost state, relative to the period before, in the CPR-only equilibrium model solved with the forecasting rule described in Section 2.4. The price clears the market each period using (17). The fixed-price regime comparison instead imposes the path in (22).

The fixed-price exercise imposes a high coffee price in the frost state and the two subsequent periods. Figure 4 shows a similar price response when the price is determined in equilibrium: a frost raises the price by about 12 percent initially and decays to zero over three years. A higher price offsets part of the revenue loss for farmers with surviving production and relaxes the CPR borrowing limit, so it widens the gap between affected and unaffected farmers.

The model-implied event studies imply the empirical prediction that motivates our design: after the shock, farmers entering with low coverage reduce debt and investment, whereas farmers entering with high coverage finance capital replacement. In the data, observed pre-shock insurance status supplies the corresponding sample split. We next estimate these conditional dynamic responses using granular variation in frost damage across nearby farms.

3 Setting and Data

Firms demand insurance due to revenue volatility and the desire to hedge against risk, which can directly arise from preferences or indirectly arise from the cost of default. While price insurance exists for commodities through derivatives markets, there are no available

revenue insurance contracts in most sectors. This could be due to a combination of unraveling induced by adverse selection or moral hazard, credit constraints to pay the insurance premium upfront, and limited commitment by insurers to truthfully fulfill claims. In many countries, including the US and our setting in Brazil, the government offers insurance not only for farmers' revenues, but also for yields (quantities) and costs (input expenditures). Two programs provide crop insurance in Brazil. PROAGRO, administered by the Central Bank and recorded in SICOR, reduces rural-credit obligations when natural events destroy the financed crop. The Rural Insurance Premium Subsidy Program (PSR), administered by the Ministry of Agriculture (MAPA), subsidizes premiums on private yield, cost, and revenue policies. Despite the widespread provision of subsidies, with insurance premiums an average of 30% below actuarially fair levels in Brazil, insurance take-up is as low as 20% for crops like coffee. While salience and search costs may be large enough frictions to result in low take-up among small "household" farmers, low take-up is common even among large "industrial" farmers in Brazil.

In addition, agriculture is a setting where liquidity shocks are particularly salient because farmers receive most of their income around harvest time. A shock that occurs between harvests can leave farmers in a difficult situation: current and future income decrease, but current expenditure increases in response to the shock. Even if the farmers have sufficient assets, they can face a cash flow shortage. This suggests that farmers' insurance take-up decisions are inherently linked to their credit access and borrowing constraints.

3.1 Data

There are three components to our main dataset: the rural credit registry, crop insurance, and other financial data. The Rural Credit Registry (SICOR) provides contract-level data for agricultural credit and the linked public crop insurance program. The contracts list the crop, the area, the expected production, the actual production, the insurance premium, the insured value, the coverage level, and the precise coordinates of the farm. We observe insurance claims that quantify the amount of the loss, the amount paid to farmers, and a description of the reason for the insurance payout. We supplement the credit and insurance data from SICOR with data from the Ministry of Agriculture (MAPA) on insurance contracts from the main private crop insurers. In MAPA, we observe similar variables to SICOR for premiums, claims, and farm characteristics.

For firms' supply chain linkages and borrowing from banks, we merge transaction-level datasets of interfirm payments, invoices, and credit operations from the BCB. The BCB credit registry (SCR) contains all firms and individuals with total debt since June 2016 of over 200 BRL, equal to around 40 USD. Using the crosswalk between the BCB datasets and the administrative registry from the federal revenue service, we observe each

counterparty’s municipality and 7-digit CNAE code, which is similar in specificity to a 6-digit Harmonized System (HS) code. We classify 16 CNAE codes as upstream of coffee farming, based on their descriptions and payment shares (Appendix B).

The SCR also reports the guarantee associated with each credit operation. We use guarantees observed before the frost and allocate an operation’s balance across guarantee classes in proportion to reported guarantee values when an operation has multiple guarantees. We distinguish guarantees backed by harvest revenue, including CPR and *penhor*, from guarantees backed by real estate. Other guarantees and records that cannot be classified remain separate rather than being assigned to either group. We aggregate these balances to construct the share of each farmer’s debt backed by land and construct the same measure within each farmer–lender relationship. Our benchmark uses the June 2021 stock, before the frost, and we use the average over the preceding twelve months as a robustness measure.²

²The classification uses the complete guarantee code rather than the first digits of the code. Guarantee choice is predetermined with respect to the frost but is not randomly assigned, so the land-share interaction is interpreted as a model-based test of heterogeneity rather than as the causal effect of choosing a land guarantee.

Table 5: Summary Statistics on the Frost Shock, Coffee Production, and Hedging

	Mean	25th Percentile	Median	75th Percentile	Number of Coffee Farmers
Farm Area (Ha)					
Insured	42.61	6.56	14.22	34.9	20,472
Uninsured	23.88	2.62	5.45	14.08	80,376
Revenue Share from Coffee (%)					
Insured	92%	97%	100%	100%	20,472
Uninsured	91%	100%	100%	100%	80,376
Frost Shock Magnitude (Damage Share)					
Insured	26%	-	-	-	20,472
Uninsured	27%	-	-	-	80,376
Insured Value (Thousand USD)					
Insured	122	25	49	125	20,472
Uninsured	-	-	-	-	80,376
Claim Payout (Share of Total Insured Value)					
Insured	30%	5%	20%	46%	20,472
Uninsured	-	-	-	-	80,376
Outstanding Debt (Thousand USD)					
Insured	80	0	45	112	20,472
Uninsured	31	0	5.6	32	80,376
Default Rate (Share of Debt Over 90 Days Overdue)					
Insured	2.5%	-	-	-	20,472
Uninsured	6.2%	-	-	-	80,376
Interest Rate (per Annum)					
Insured	5.6%	1.1%	4.4%	6.5%	15,747
Uninsured	9.5%	2%	4.2%	5.5%	60,760
Outstanding Debt Maturity (Years)					
Insured	7.5	5.9	7.3	8.8	15,747
Uninsured	8.1	5.9	8.8	9.9	60,760
Annual Payment Outflows (Thousand USD)					
Insured	743	11	43	191	20,472
Uninsured	275	1.6	5.58	18.6	80,376
Agricultural Inputs Payments (Share of Total Payments)					
Insured	48%	20%	48%	75%	20,472
Uninsured	45%	11%	45%	78%	80,376

Notes: These summary statistics use data from the BCB and SICOR. For insured farmers, frost severity is measured from the verified claim relative to insured value. For farmers without an own-policy loss measure, frost severity is inferred from the inverse-distance-weighted damage surface described in equation (27). Insured value and claim payouts are unavailable for uninsured farmers and are shown as dashes. The interest rates are low because the government heavily subsidizes agricultural lending in Brazil.

Table 5 shows summary statistics from our main dataset. Insured and uninsured farmers differ along most dimensions. Uninsured farmers pay higher interest rates, default at higher rates, have smaller farms, and receive less revenue.³ However, insured and uninsured farmers were similarly affected by the frost shock.⁴

³We define default the standard way in Brazil: debt that is unpaid at least 90 days after the due date.

⁴There is little correlation between the frost shock magnitude and any systematic differences in insurance

A key challenge in constructing the combined dataset of contracts at the farmer level was the creation of a crosswalk across insurance and credit contracts that small-scale farmers took under their personal identifier rather than their firm identifier. We believe that we are among the first to construct such a crosswalk for all farmers in any country.

For municipality-level statistics on weather and farm output that we only feature in Appendix A, we merge weather station data from Brazil’s National Institute of Meteorology to agricultural survey data from the Municipal Agricultural Production (PAM) and Systematic Survey of Agricultural Production (LSPA) from the Brazilian Institute of Geography and Statistics (IBGE).

3.2 Frost Shock

We study the July 2021 frost in Brazil, which unexpectedly destroyed part of the capital stock for affected coffee farmers. Three severe waves of frost affected the states of São Paulo, Minas Gerais, and Paraná between the end of June 2021 and the beginning of August 2021. The frost was the worst in decades. Due to timing and geography, the frost primarily affected perennial crops like oranges and coffee, rather than annual crops like soybeans and maize. Because the frost arrived after most of the 2021 coffee harvest, harvest data record little of the damage.⁵ Measuring frost with weather data can be imprecise because small differences in geography and soil, in combination with small differences in temperature, can correspond to large differences in losses.⁶

We interpret the frost as a capital-stock shock, rather than solely a productivity or cash-flow shock, because coffee is a perennial crop. The frost killed coffee plants, whose value is included in the collateral that farmers post for loans. The timing of the frost, at the end of the harvest season, meant that the 2021 coffee harvest was not heavily disrupted. We measure plot-level damage as the frost indemnity divided by insured value. For yield policies, this ratio is zero for yield losses below the 30 or 35 percent deductible, reaching one only at total loss, so it understates physical damage on partially damaged plots. Claims are paid only after an agronomist visits the plot and validates the loss. We observe the cause of loss and the coordinates of each insured plot. This produces a granular damage measure that captures agronomic differences missed by municipality-level weather or harvest outcomes.⁷

take-up across the country or within small geographical regions.

⁵Indeed, the challenge of estimating the effect of frost without detailed microdata at the coffee farm level has been recognized for a long time. For example, see Stevens (1955) for a historical discussion.

⁶See Appendix A for more details about the measurement of the frost shock and the geographical distribution of crops in Brazil.

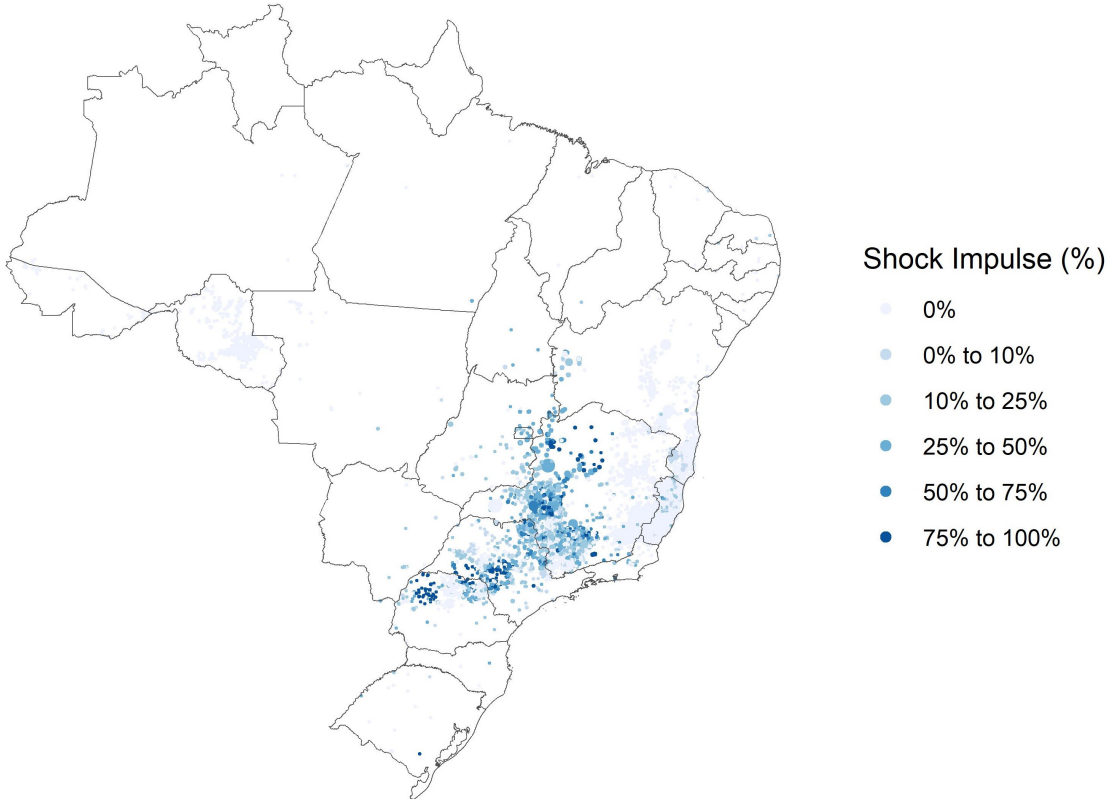
⁷Frost events are rare but consequential for Brazilian coffee production. At least 17 frost events occurred during the twentieth century.

For a farmer with a verified own-policy loss, the damage measure is the frost claim divided by insured value. For a farmer without an own-policy loss measure, we infer damage from nearby insured coffee plots. We observe coordinates x_i and, for each insured neighbor $j \in \mathcal{N}_i$, the ratio I_j of the frost claim to insured value. The spatial prediction is

$$\widehat{FS}_i = \frac{\sum_{j \in \mathcal{N}_i} w_{ij}(x_i) I_j}{\sum_{j \in \mathcal{N}_i} w_{ij}(x_i)}, \quad w_{ij}(x_i) = \frac{1}{d(x_i, x_j)^\beta}, \quad (27)$$

where $d(x_i, x_j)$ is the distance between farms and $\mathcal{N}_i$ contains the 52 nearest insured plots. The benchmark uses inverse-distance weights ($\beta = 1$). We rescale positive predictions for farmers without insurance to the mean and dispersion of verified losses, so a percentage point has the same interpretation in the insured and uninsured subsamples. The resulting FS_i is defined for every coffee farmer and is fixed before we merge post-frost credit and payment outcomes.

Figure 5: Frost Shock Impulse Based on Coffee Insurance Claims



Notes: The insurance data are from SICOR. Each dot represents a farm that grew coffee in 2021. The size of the dot scales in proportion to the size of the farm. The color of the dot represents the shock impulse, where darker colors represent higher magnitudes. The coffee-growing regions that were unaffected by the frost shock, with shock impulse equal to 0%, were Rondônia, eastern Minas Gerais, Espírito Santo, and eastern Bahia; these regions can be seen clearly in Figure A1 in Appendix A.

Figure 5 shows the geographical distribution of the frost shock. Although some coffee-growing regions (e.g. northern Paraná through northwestern Minas Gerais) were affected more than others (e.g. eastern Minas Gerais through Bahia),⁸ there was variation in the severity of the frost shock even within each municipality, an administrative division that is on average half the size of a county in the contiguous US.

4 Empirical Methodology and Results

The model event study motivates our empirical specification. In the model, we compare farmers who receive the capital shock with farmers in the same initial coverage group who do not. This grouping is the intensive-margin analogue of, not a structural mapping to, observed insurance take-up. In the data, we replace the simulated shock indicator with the continuous farm-level measure of frost damage and estimate responses separately for farmers with and without observed insurance contracts. We do not compare insured farmers directly with uninsured farmers because insurance status is not randomly assigned.

Our identifying assumption is that the granular impulse of the frost was as good as random, given the outcome-specific fixed effects stated below. Our justification is that the frost was not anticipated at the time of planting, the most severe since the advent of modern weather stations, and the weather forecast did not predict it until a week beforehand, after all planting and insurance decisions were sunk. Even then, forecasts did not capture the granular spatial variation in the realized shock.

Our regressions encompass three sets of outcome variables. The credit regressions examine affected farmers' outcomes $y_{i\ell jt}$, such as outstanding debt and maturity, with separate event-study coefficients by whether the farmers had existing insurance contracts in place. Let i be farmer, ℓ lender, j municipality, g insured versus uninsured, and t quarter. Let $s_{ij\tau}$ interact our preferred shock metric FS_i , defined above, with an indicator for calendar quarter τ : $s_{ij\tau} = FS_i \mathbb{1}\{t = \tau\}$. Let Ins_i indicate insurance in place at the beginning of the frost growing season.

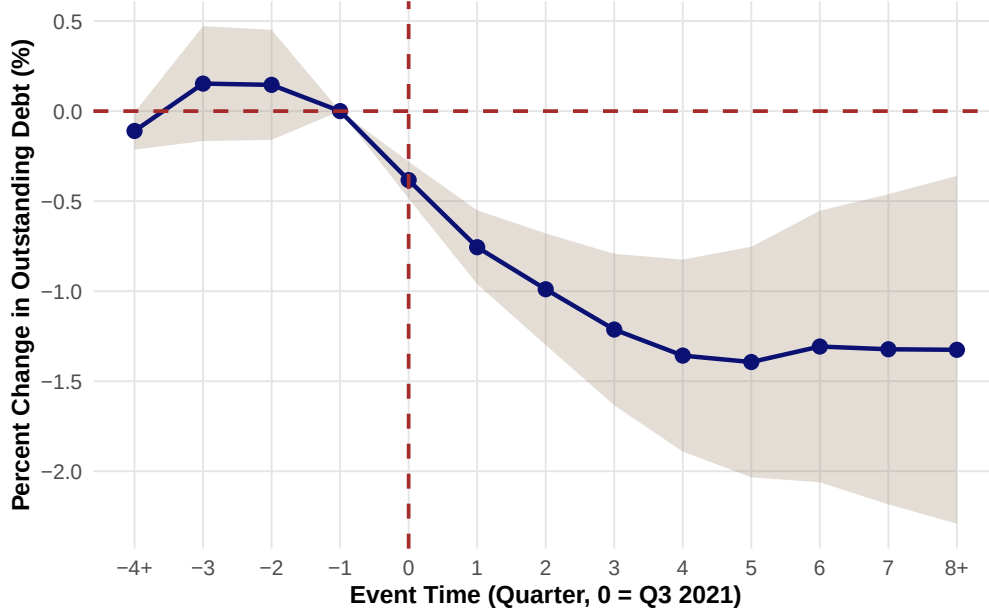
$$y_{i\ell jt} = \sum_{\tau=2019Q1}^{2023Q4} \beta_{\tau}^C s_{ij\tau} Ins_i + \sum_{\tau=2019Q1}^{2023Q4} \gamma_{\tau}^C s_{ij\tau} No_Ins_i + \alpha_{i\ell} + \alpha_{\ell jt} + \epsilon_{i\ell jt}. \quad (28)$$

Farmer–lender fixed effects absorb persistent match characteristics, while lender–municipality–quarter fixed effects absorb local changes in lender supply. The omitted period is 2021Q2, immediately before the frost, and standard errors are clustered by farmer. The β_{τ}^C coefficients compare more and less damaged insured farmers, and the γ_{τ}^C coefficients make the same comparison among uninsured farmers. Conditional on the fixed effects, the identi-

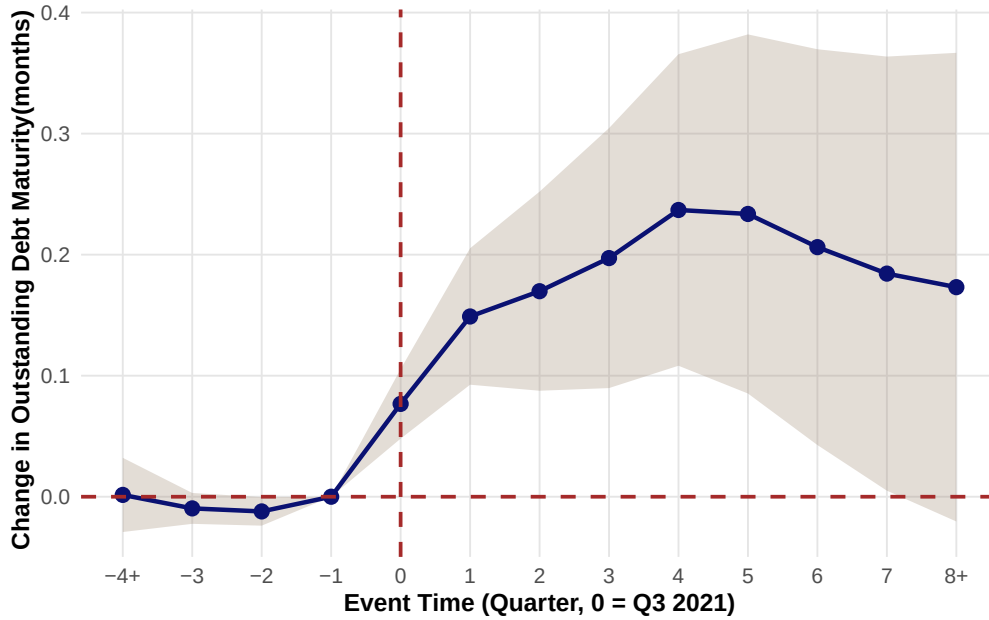
⁸See Figure A1 in Appendix A for a map of coffee production in Brazil.

fying variation is the granular spatial variation in frost damage.

Figure 6: Credit Regression Results of $\{\gamma_{\tau}^C\}$ for Uninsured Farmers



(a) Outstanding debt decreased



(b) Debt maturity increased

Notes: These regressions use BCB credit-registry data from 2019Q1 through 2023Q4. The coefficients $\{\gamma_{\tau}^C\}$ in equation (28) compare more and less damaged uninsured farmers, relative to 2021Q2. Shock magnitude is the frost indemnity as a percentage of insured value. For uninsured farmers, it is the rescaled spatial prediction in equation (27). Outcomes are outstanding debt in panel (a) and outstanding debt maturity in panel (b). All specifications include farmer–lender and lender–municipality–quarter fixed effects. Bars report 95% confidence intervals based on standard errors clustered by farmer.

Figure 6 shows the event study coefficients for the $\{\gamma_\tau^C\}$ coefficients, across event time τ on the horizontal axis, for the outcome y_{ijt} outstanding debt balance in Figure 6a and the outcome y_{ijt} debt maturity in Figure 6b.⁹ The takeaway from Figure 6 is that each additional percentage point of coffee plant damage for uninsured farmers, comparing against uninsured farmers with zero damage, corresponds to a 0.4% immediate reduction and 1.4% long-run reduction in outstanding debt balance. Our interpretation is that farmers' net worth decreased, so their borrowing constraints tightened and reduced their debt load despite having higher marginal returns to capital. The takeaway from Figure 6b is that each additional percentage point of coffee plant damage for uninsured farmers, comparing against uninsured farmers with zero damage, corresponds to a 0.08 month immediate increase and 0.2 month long-run increase in outstanding debt maturity. Our interpretation is that farmers renegotiated debt and received forbearance on their loans.

By comparison, we show in Figure A4 in Appendix A that the results for $\{\beta_\tau^C\}$ in regression (28) are zero across all event time, as we expect. Insurance cushions farmers from the financial impacts of the shock. In addition, we find no significant effects on other credit outcomes: the debt interest rate, the default rate, and the composition of farmers' debt.

4.1 Collateral Composition and the Credit Response

The model predicts that the response of borrowing depends on whether credit is backed by future harvest revenue or by land. We test this prediction among uninsured farmers, for whom the shock is not offset by an insurance payout. Let L_i be the share of farmer i 's pre-shock debt backed by land and let X_i contain predetermined farm characteristics. We estimate

$$y_{ijt} = \sum_{\tau \neq 2021Q2} \beta_\tau s_{ij\tau} + \sum_{\tau \neq 2021Q2} \kappa_\tau L_i \mathbb{1}\{t = \tau\} + \sum_{\tau \neq 2021Q2} \delta_\tau s_{ij\tau} L_i + \sum_{\tau \neq 2021Q2} \mathbb{1}\{t = \tau\} X_i' \rho_\tau + \sum_{\tau \neq 2021Q2} s_{ij\tau} X_i' \zeta_\tau + \alpha_{il} + \alpha_{\ell jt} + \epsilon_{ijt}. \quad (29)$$

The outcome is log outstanding debt. Farmer–lender fixed effects absorb persistent differences across credit relationships, while lender–municipality–quarter fixed effects absorb local changes in credit supply. Standard errors are clustered by farmer. The coefficients $\{\delta_\tau\}$ measure whether more severe frost damage is followed by a different borrowing response among farmers whose debt was more intensively backed by land. The specification includes interactions of event time and frost damage with pre-shock farm size, coffee rev-

⁹See Figure A4 in Appendix A for the results for the $\{\beta_\tau^C\}$ coefficients.

enue share, debt per hectare, maturity, interest rates, undrawn credit, and collateral value. We demean these controls within the estimation sample so that $\{\beta_\tau\}$ retain the interpretation of the response at the average farmer.

We use the continuous land share because it preserves the information in mixed guarantee portfolios. We also compare farmers with at least 80% of debt backed by land to farmers with at least 80% backed by harvest revenue, replace the June 2021 share with its twelve-month pre-shock average, and estimate the interaction within farmer–lender relationships. The relationship specification includes the lower-order event-time interaction with the relationship land share. These comparisons do not assume that guarantee choice is random. Their purpose is to test the model’s prediction that the same physical shock transmits differently through different forms of pledgeable collateral.

The payment regressions examine how farmers’ purchases changed, specifically focusing on purchases from suppliers in sectors that are consistent with investment in re-growing crops or in the quality of the land.¹⁰

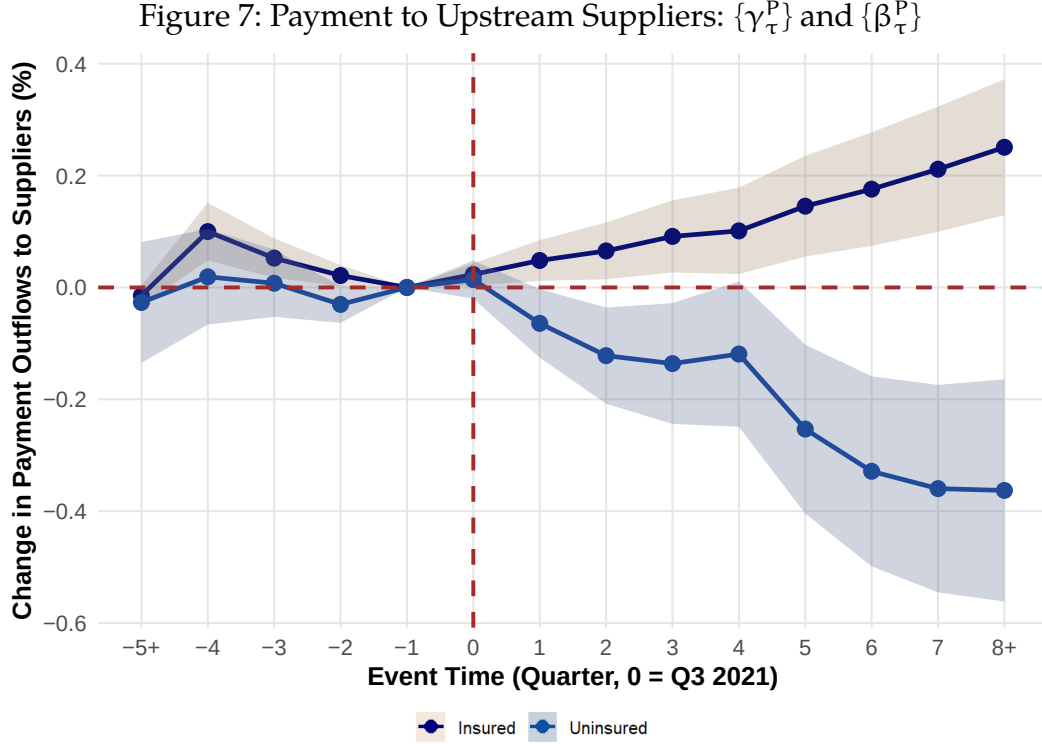
$$y_{ijst} = \sum_{\tau=2019Q1}^{2023Q4} \beta_{\tau}^P s_{ij\tau} \text{Ins}_i \text{Upstream}_s + \sum_{\tau=2019Q1}^{2023Q4} \gamma_{\tau}^P s_{ij\tau} \text{No_Ins}_i \text{Upstream}_s + \alpha_{is} + \alpha_{jmgt} + \epsilon_{ijst}, \quad (30)$$

where subscript i is farmer, j is municipality, s is supplier, t and τ are quarters, and m is the supplier’s 7-digit CNAE code. Farmer–supplier fixed effects absorb persistent trading relationships; municipality–supplier–sector–insurance–group–quarter fixed effects absorb common local and sectoral shocks. The regressions are weighted by farm area, and standard errors are clustered by municipality. The outcome y_{ijst} is log payment *outflows* to upstream suppliers of agricultural inputs and services, such as seedlings, pruning, land-preparation services, and agricultural machinery. The coefficients identify differential responses to frost damage separately by initial insurance status. We do not compare insured farmers directly with uninsured farmers due to selection issues, and we motivate the specification using our model with this in mind.

Figure 7 shows the event study coefficients for the $\{\gamma_\tau^P\}$ and $\{\beta_\tau^P\}$ coefficients from equation (30), across event time τ on the horizontal axis. For each percentage point of additional damage, more damaged uninsured farmers reduce purchases from upstream suppliers relative to less damaged uninsured farmers, with a long-run decrease of 0.38 percent. Our interpretation is that these purchases are partly financed with credit, so they fall when borrowing constraints tighten as net worth declines. By comparison, more dam-

¹⁰We do not use incoming payments to proxy for farmers’ sales because many of the farmers’ sales are sold forward by CPR, so payments received after the frost partially reflect quantities and prices contracted before the frost.

aged insured farmers increase these purchases relative to less damaged insured farmers, with a long-run increase of 0.24 percent per 1 percentage point of damage. We believe that this is due to the liquidity injection from the timing of the insurance payout, which is typically as soon as an insurance adjuster can reach the farm, versus typical farm revenues that occur months later.



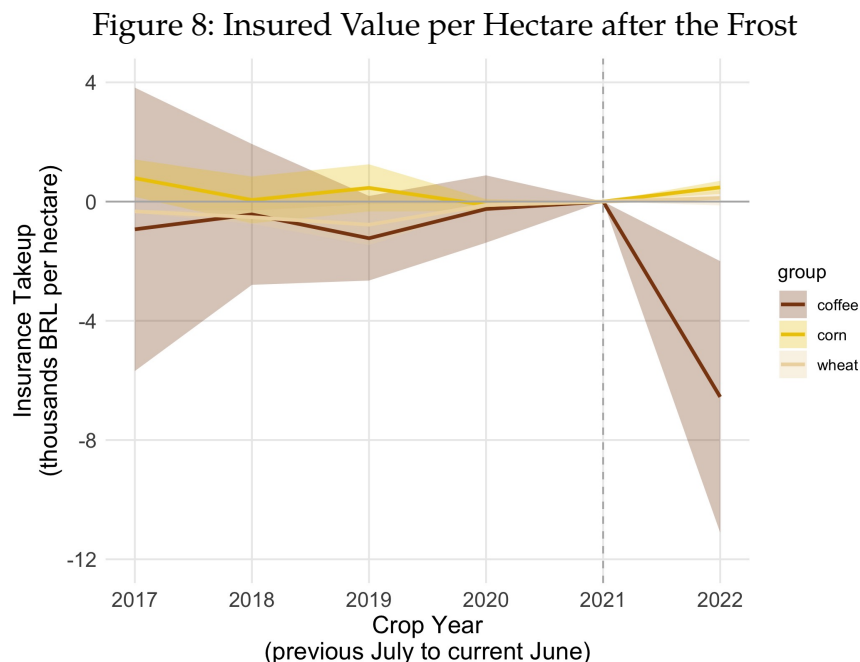
Notes: These area-weighted regressions use BCB payment data and correspond to equation (30). Coefficients $\{\gamma_\tau^P\}$ compare more and less damaged uninsured farmers. Coefficients $\{\beta_\tau^P\}$ make the same comparison among insured farmers. Shock magnitude is the frost indemnity as a percentage of insured value. For uninsured farmers, it is the rescaled spatial prediction in equation (27). All specifications include farmer–supplier and municipality–supplier–sector–insurance-group–quarter fixed effects. Bars report 95% confidence intervals based on standard errors clustered by municipality.

The insurance regression in equation (31) examines the intensive margin of coverage among observed policies. The outcome is insured value per hectare:

$$y_{ijct} = \sum_{\tau=2017}^{2022} \beta_{c\tau}^I FS_{ic} \mathbb{1}\{t = \tau\} + \eta \text{ExpectedProduction}_{ict} + \alpha_i + \alpha_{jct} + \epsilon_{ijct}, \quad (31)$$

where i is farmer, c crop, j municipality, and t growing season. Farmer fixed effects absorb permanent differences across policyholders, and municipality–crop–year fixed effects absorb local changes in insurance supply and crop conditions. Standard errors are clustered by farmer. The coefficients $\beta_{c\tau}^I$ compare insured value per hectare across more and less

damaged plots among observed policies. They do not estimate entry into insurance among previously uninsured farmers.



Notes: These regressions use SICOR policy data and correspond to equation (31) for coffee, corn, and wheat. The outcome is insured value per hectare. Coefficients compare observed policies on more and less damaged plots relative to the 2021 growing season. All specifications include farmer and municipality–crop–year fixed effects and control for expected production. The shaded area shows 95% confidence intervals based on standard errors clustered by farmer. The post-frost relationship is concentrated in coffee, consistent with persistent destruction of productive capital in a perennial crop.

Figure 8 shows that insured value per hectare falls with frost severity for coffee after the frost. The corresponding relationship is absent for corn and wheat, annual crops that are replanted each season even though their insured losses were substantial.¹¹ This contrast is consistent with the model’s central net-worth mechanism: persistent destruction of productive capital tightens financing capacity and increases the opportunity cost of state-contingent protection.

In Appendix A, we show additional results that complement the main results: we find no significant differences in default rates in Figure A6, interest rates in Figure A5, or insurance premiums in Figure A7 between affected and unaffected farmers, in both the insured and uninsured groups. We interpret these findings as consistent with a setting where the financial market anticipates farmers’ ability to repay debt in the event of shocks and allows for some form of state-contingent repayment. This may occur through mechanisms such as mandatory insurance or automatic debt renegotiation, as observed in certain rural

¹¹Soybeans were unaffected because the frost occurred outside the soybean growing season.

credit contracts. However, this does not imply complete risk-sharing. As the model shows, limited commitment constrains both insurance and credit uptake.

The empirical results have a common pattern. More severely damaged uninsured farmers reduce outstanding debt and purchases from upstream input suppliers, while debt maturity increases. More severely damaged insured farmers instead increase purchases from those suppliers. Among observed policies, insured value per hectare falls with damage for coffee but not for annual crops. Interest rates, default, and insurance premiums do not respond significantly. Together, these estimates show adjustment through quantities and maturity rather than prices or default.

These results follow the model event-study predictions. Uninsured farmers reduce debt and investment even though the shock increases their marginal return to replacing capital, while insured farmers invest to rebuild. The debt-maturity response also shows that state contingency can arise through renegotiation even when observed interest rates and default do not change.

5 Policy Implications

In this section, we use the CPR-and-land collateral structure, setting $\lambda_K = 0$, to derive policy implications.

5.1 Emergency Credit Lines

We examine the effectiveness of emergency credit lines within the model's framework. Following the frost shock, the Central Bank of Brazil allocated 1.3 billion BRL (approximately 250 million USD) to be disbursed as credit lines to coffee farmers impacted by the event. Financial intermediaries, such as banks and credit cooperatives, were responsible for disbursing the funds. However, only around half of the total amount was actually lent, and according to credit registry microdata, a disproportionately high share went to farmers already holding some form of quantity insurance: 40% of all lending went to the 20% of farmers who had purchased insurance.¹² Given the extent of the frost shock, why did the emergency credit not reach those who needed it most?

The model highlights a key channel: an emergency credit line does not increase borrowing capacity when it is subject to the same collateral constraint as ordinary rural credit. Let L_i denote the emergency loan disbursed today, let $R_L L_i$ be its promised repayment next

¹²For the total amount allocated, see Resolution of the National Monetary Council No. 4,954 of October 21, 2021. For the total amount disbursed, see a report from the Ministry of Agriculture.

period, and let $\bar{L}_i$ be the loan limit. The current budget becomes

$$c_i = W_i + \mathbb{E} \left[\frac{b_i(S', z')}{R} \right] + L_i - k_i. \quad (32)$$

If ordinary and emergency credit use the same collateral, then

$$b_i(S', z') + R_L L_i \leq \lambda p(S') A_i \theta_i(S', z')^\alpha k_i^\alpha + \chi_i q_i(S', z'), \quad 0 \leq L_i \leq \bar{L}_i, \quad \forall (S', z'). \quad (33)$$

Increasing the announced limit $\bar{L}_i$ does not relax a collateral constraint that already binds. This conclusion holds when the emergency rate equals the ordinary rural rate. If $R_L < R$, a unit of collateral supports $1/R_L$ rather than $1/R$ in current funds, so a subsidized line expands borrowing capacity, but only in proportion to the farmer's remaining collateral. The expansion is smallest for the farmers whose collateral the frost destroyed. As a result, the additional funds might not be disbursed or flow toward insured and otherwise less constrained farmers. Emergency credit cannot replace insurance when both depend on the same pledgeable collateral.

A government guarantee changes this result. Suppose that the guarantee covers fraction ϕ_G of the emergency loan's promised repayment. The borrowing constraint becomes

$$\begin{aligned} b_i(S', z') + (1 - \phi_G) R_L L_i &\leq \lambda p(S') A_i \theta_i(S', z')^\alpha k_i^\alpha \\ &\quad + \chi_i q_i(S', z'), \\ 0 \leq L_i \leq \bar{L}_i, \quad \phi_G &\in [0, 1]. \end{aligned} \quad (34)$$

Only the non-guaranteed share $(1 - \phi_G) R_L L_i$ uses the farmer's collateral capacity. In the model, increasing the effective guarantee ϕ_G , rather than the announced volume alone, relaxes the farmer's collateral constraint. Other policies can relax the two collateral constraints directly. Better registration, verification, or monitoring of CPR proceeds raises harvest pledgeability λ , while better land titles, registries, or enforcement raise land pledgeability χ .

The low disbursement rate is consistent with the model because emergency credit is least likely to reach constrained farmers when it has the same collateral requirements as ordinary rural credit.

These results are related to the experience of the Paycheck Protection Program in the United States. Joaquim and Wang (2022) show that emergency lending through financial intermediaries did not always reach the firms most affected by the COVID crisis. In both settings, the announced availability of funds can differ from the ability of constrained firms to use them. Corporate emergency credit following the floods in Rio Grande do Sul provides another setting in which the model's distinction between announced liquidity

and usable collateral capacity can be examined. We do not use that episode as evidence for the mechanism in this paper.

5.2 Increasing the Pledgeability of Collateral

The distinction between liquidity and pledgeability also clarifies the role of recent innovations in agricultural credit. Registration of CPR and CPR-F contracts, together with satellite or drone verification of crop conditions and harvest proceeds, can increase the fraction of future revenue that lenders can verify and recover. In the model, this is an increase in λ . Monitoring alone is not sufficient. The proceeds must be identifiable, diversion to unregistered buyers must be limited, and the lender must have an enforceable claim.

Better land titles and registries instead increase χ by making a larger share of land value available as collateral. Fractionalization or tokenization can have the same effect only if the registry prevents multiple pledges and establishes the priority of each lender's claim. It may also allow a farmer to pledge unused land value to another lender, although the resulting effect on lender competition is outside the model because the interest rate is exogenous.

5.3 Climate Stress Testing and the Incidence of Risk

The model also changes how we interpret climate stress tests. These exercises normally ask how much climate risk is borne by banks and insurers. In our setting, low financial-sector exposure need not mean that the underlying risk is small. When collateral constraints prevent farmers from borrowing or purchasing insurance, the risk remains with the farmers. They respond by reducing investment, which delays the recovery of coffee production. Under the maintained inelastic-demand benchmark, the reduction in aggregate supply raises the coffee price and passes part of the loss to consumers. The price increase also raises the value of surviving harvest and grove collateral, so the final effect on borrowing capacity depends on which farmers retain productive capital after the shock.

This is the other side of climate stress testing. A stress test that reports only the direct losses of financial institutions can miss the losses borne by farmers, suppliers, and consumers when credit and insurance are rationed.

6 Conclusion

Limited collateral pledgeability provides a common explanation for three features of climate adaptation that are usually studied separately: producers under-insure before a dis-

aster, post-disaster credit fails to reach the producers who were damaged most, and physical shocks propagate through supply chains. Insurance and production compete for the same borrowing capacity, so under-insurance is a constrained-optimal choice rather than a behavioral mistake. After a shock, uninsured farmers can have the highest marginal return to replacing capital but the least ability to finance it.

We extend the model to distinguish future harvest revenue pledged through a CPR from land collateral. Bare land can cushion a frost shock, whereas the part of land value that reflects standing coffee trees remains exposed to physical destruction. The same frost that destroys the asset raises the price of what survives, so the price increase affects both current revenue and the value of price-sensitive collateral. The equilibrium price operates through both a budget channel and a collateral channel, and an aggregate shock widens the gap between damaged and undamaged farmers rather than only shifting the level of borrowing.

The empirical results follow the model predictions. More severely damaged uninsured coffee farmers reduce borrowing and input purchases and extend debt maturity. More severely damaged insured farmers instead increase input purchases to replace damaged capital. Among observed policies, insured value per hectare falls with damage for coffee but not for annual crops. We find no corresponding increases in interest rates, default, or insurance premiums. The reduction in purchases from upstream sectors shows how under-insurance propagates a physical climate shock through the supply chain. The credit channel alone was insufficient to compensate for the lack of insurance.

These findings change what climate policy should target. Increasing the supply of intermediated credit is not equivalent to increasing the borrowing capacity of the farmers most affected by the shock, which helps explain why around half of Brazil's emergency credit line was disbursed and why a disproportionate share of lending reached farmers who were already insured. Public guarantees and measures that raise the pledgeability of harvest proceeds and land act on the constraint that binds. The same point applies to climate stress testing. Low exposure of financial institutions can reflect an inability of farmers to transfer risk rather than resilience, leaving more of the loss with farmers, suppliers, and consumers through lower investment, lower output, and higher coffee prices.

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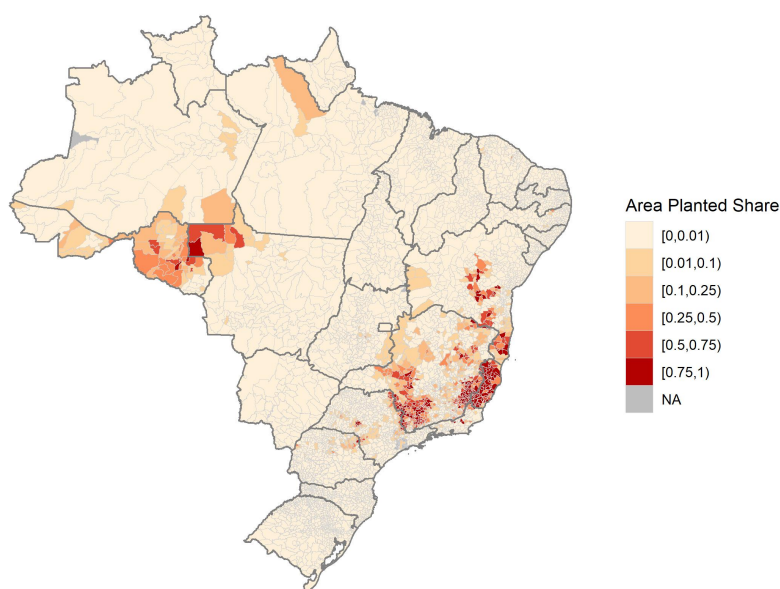
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A Empirical Appendix

The purpose of this section is to provide additional context on coffee farming in Brazil and on the frost shock. Coffee is a perennial crop whose coffee beans are harvested from coffee cherries that typically grow on the plant each year. Coffee plants can live for many decades, and typically do not begin producing cherries until the third growing season. The growing season of coffee in south central Brazil begins in September to November, with harvest in June through August. Coffee plants are sensitive to cold temperatures, with low frost tolerance. The primary coffee growing regions in Brazil, depicted in red in Figure A1, are in regions where nighttime winter temperatures rarely drop below 10 Celsius.

Figure A1: Geographical Distribution of Coffee Production in Brazil in 2021

Coffee Area Planted Share of Agriculture in 2021, by Municipality



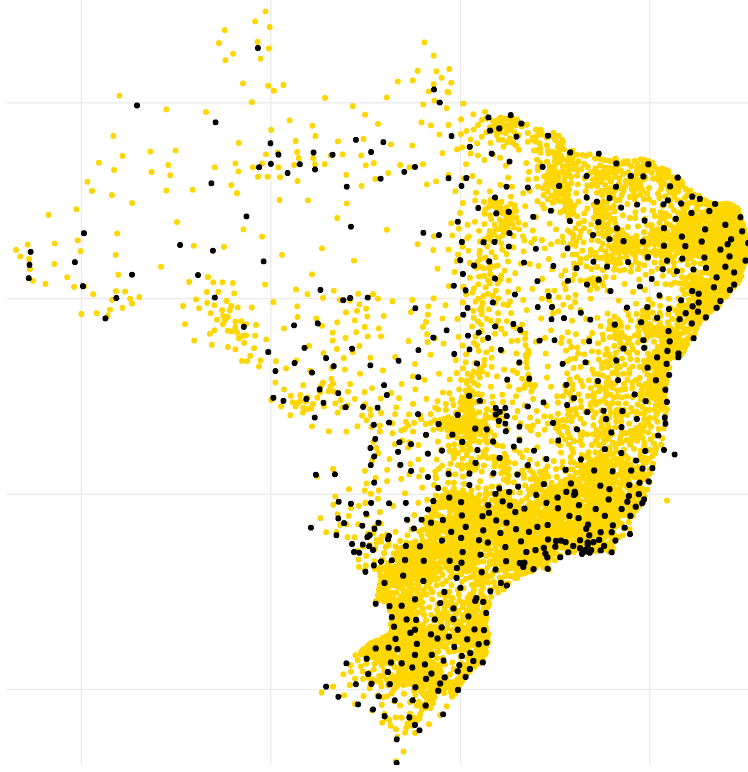
Source: Produção Agrícola Municipal

Notes: This figure uses data from the Municipal Agricultural Production (PAM) survey to plot the municipality-level share of agricultural land in 2021 where coffee was planted. In Brazil, the municipality is the administrative division that is roughly analogous to the county in the US.

However, during the frost shock, nighttime temperatures were lower than climatic averages by more than 10 degrees Celsius for multiple periods of consecutive days. The first frost did not appear in weather forecasts until 3 days before the frost on June 28, 2021, and anecdotes from news articles suggest that farmers had limited options to protect their trees, given the scale of coffee farming and the fragility of coffee plants. To assign the distribution of weather variables to each municipality, we follow the methodology of

Deschênes and Greenstone (2007) and subsequent papers in environmental economics.

Figure A2: Locations of Weather Stations in Brazil in July 2021



Notes: This figure shows the locations of weather stations from Brazil's National Institute of Meteorology in yellow, as well as municipality centroids in black.

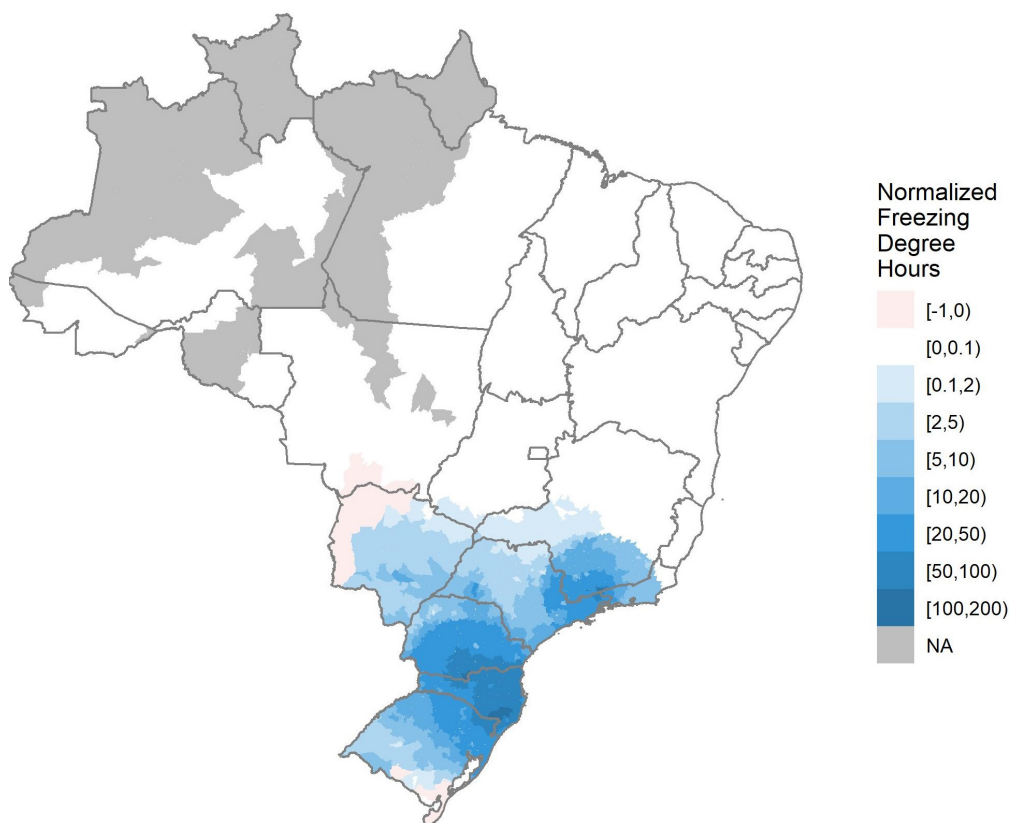
Using station-by-hour data from weather stations monitored by Brazil's National Institute of Meteorology, whose distribution is shown in Figure A2, we count the number of hours over the frost shock period of June 28 through August 2, 2021 in each temperature bin of width one degree Celsius. Then, we use inverse distance weights $\omega_{s,i}$ between municipality i centroids and weather stations s with a 100 kilometer radius to define the municipality-level count of hours in each temperature bin. We define freezing-degree-hours (FDH) similar to growing-degree-days:

$$FDH_i = \sum_{s \text{ near } i} \omega_{s,i} \sum_{\text{hour } h} \max\{0, -T_{s,h}\}. \quad (35)$$

where $T_{s,h}$ is mean hourly temperature at weather station s . Figure A3 shows the distribution of FDH_i across municipalities i with sufficient weather data. However, the sensitivity of coffee plants to cold weather is highly non-linear, which a quasi-linear variable like FDH cannot accurately capture. Coffee plants begin to experience defoliation and frost burn at temperatures as high as 5 Celsius, which a much wider swath of central Brazil

experienced during the frost shock. The extent of damage varies with wind, and even the nutrient composition of the soil, in addition to temperature. As a result, we believe that Figure 5 captures the impulse of the frost shock more accurately than any weather-based metric, like Figure A3, that follows the standard methods from the environmental economics literature or the climate finance literature.

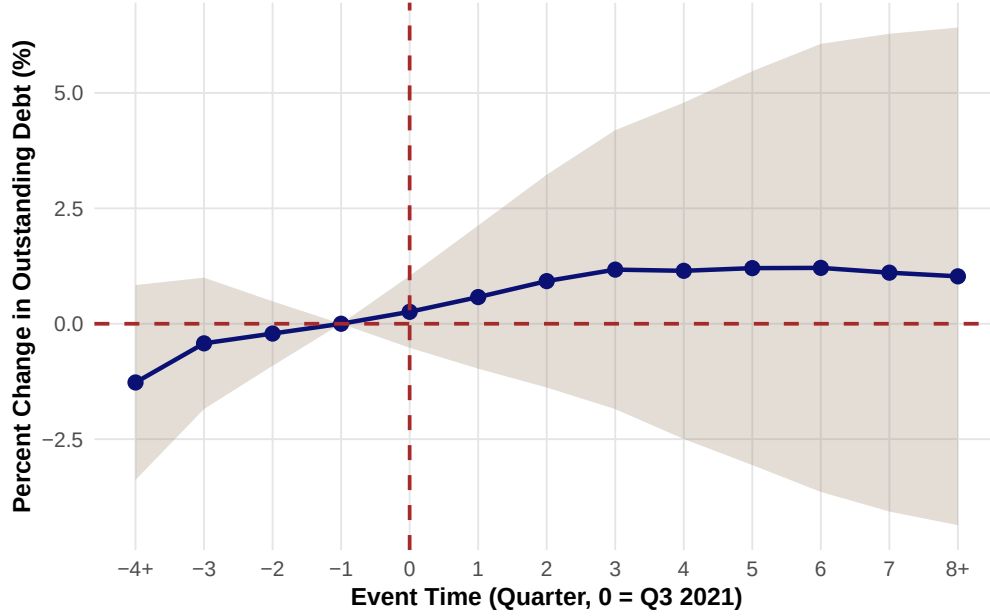
Figure A3: Municipality-Level Freezing-Degree-Hours During the Frost Shock Period



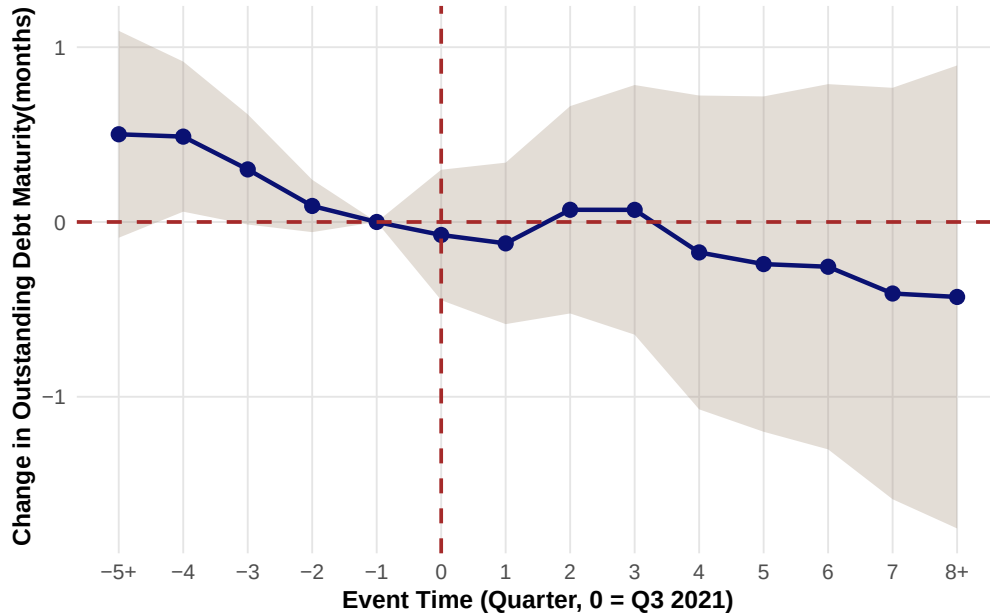
Notes: This figure shows the geographical distribution of freezing-degree-hours (FDH), normalized by the 2011-2020 average, in each municipality. We compute FDH following equation (35) using weather station data from Brazil's National Institute of Meteorology.

A.1 Additional Event Studies Results

Figure A4: Credit Regression Results of $\{\beta_{\tau}^C\}$ for Insured Farmers



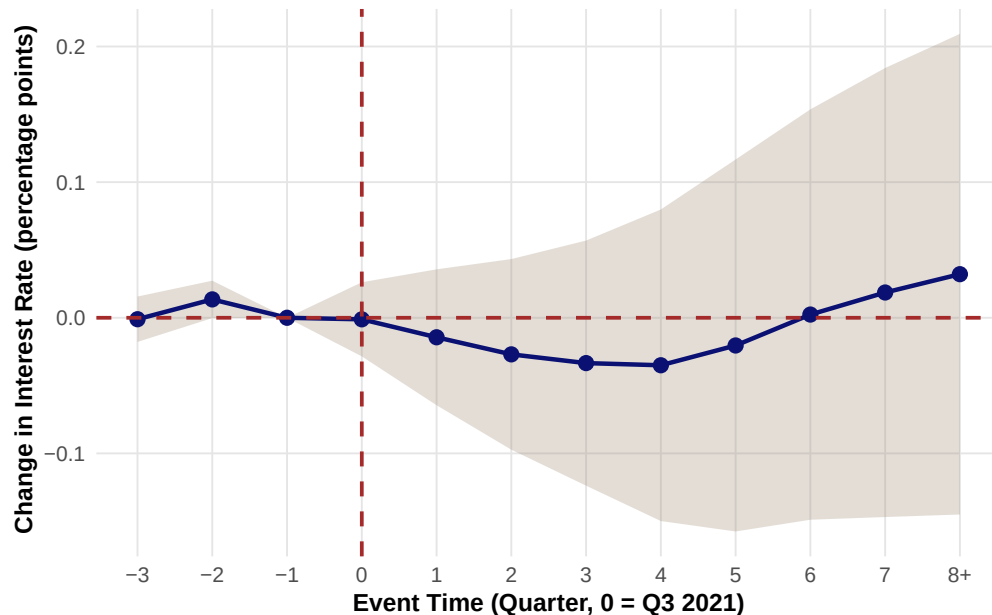
(a) No statistically significant effects on outstanding debt



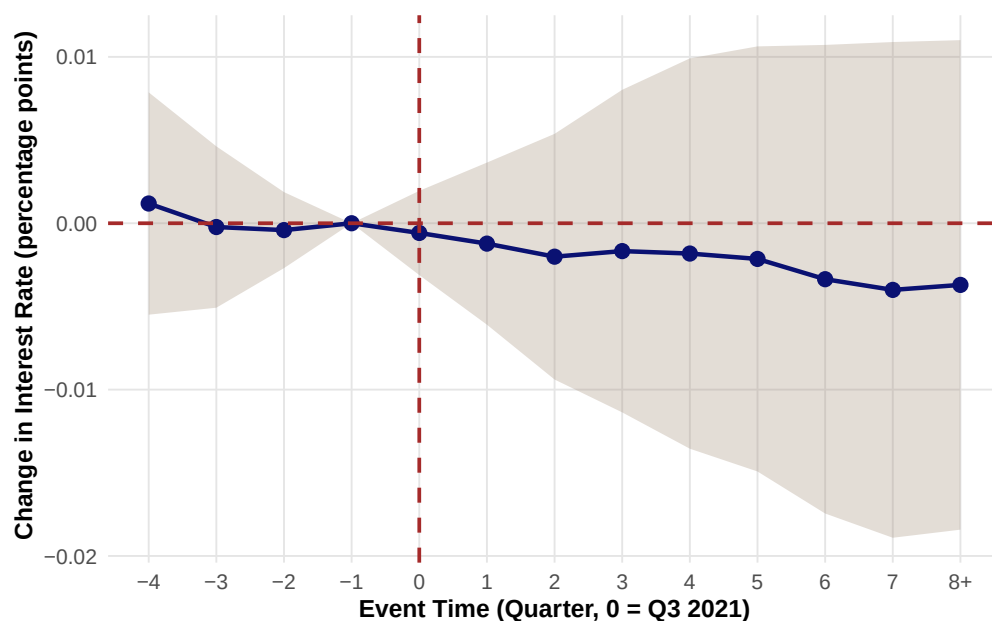
(b) No statistically significant effects on debt maturity

Notes: These BCB credit-registry regressions report $\{\beta_{\tau}^C\}$ from equation (28), comparing more and less damaged insured farmers relative to 2021Q2. Shock magnitude is the frost indemnity as a percentage of insured value. For uninsured farmers, it is the rescaled spatial prediction in equation (27). Outcomes are outstanding debt in panel (a) and debt maturity in panel (b). Specifications include farmer–lender and lender–municipality–quarter fixed effects. Bars report 95% confidence intervals based on standard errors clustered by farmer.

Figure A5: Interest Rate Regression Results for Uninsured Farmers and Insured Farmers



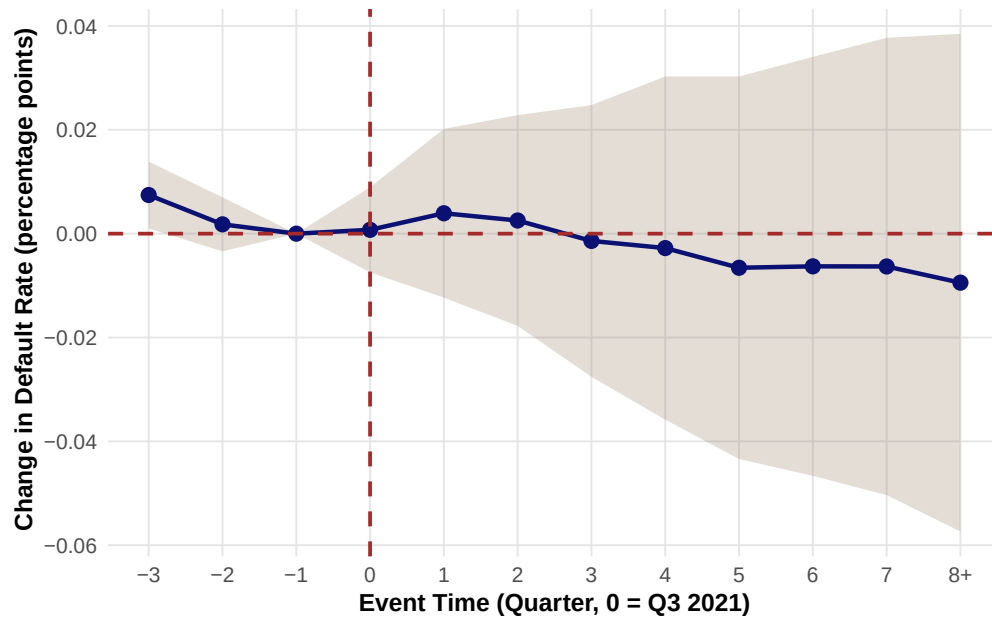
(a) No statistically significant effects on uninsured farmers' debt interest rate.



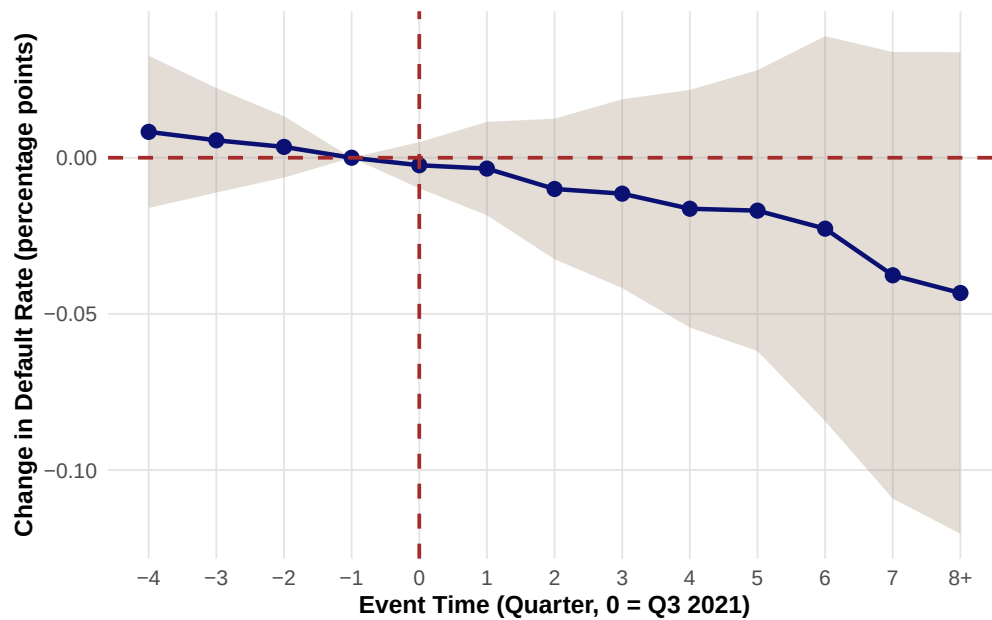
(b) No statistically significant effects on insured farmers' debt interest rate.

Notes: These BCB credit-registry regressions compare more and less damaged farmers relative to 2021Q2. The outcome is the outstanding-balance-weighted average annual interest rate. Panel (a) reports uninsured farmers and panel (b) insured farmers. Specifications include farmer–lender and lender–municipality–quarter fixed effects. Bars report 95% confidence intervals based on standard errors clustered by farmer.

Figure A6: Default Rate Results for Uninsured Farmers and Insured Farmers



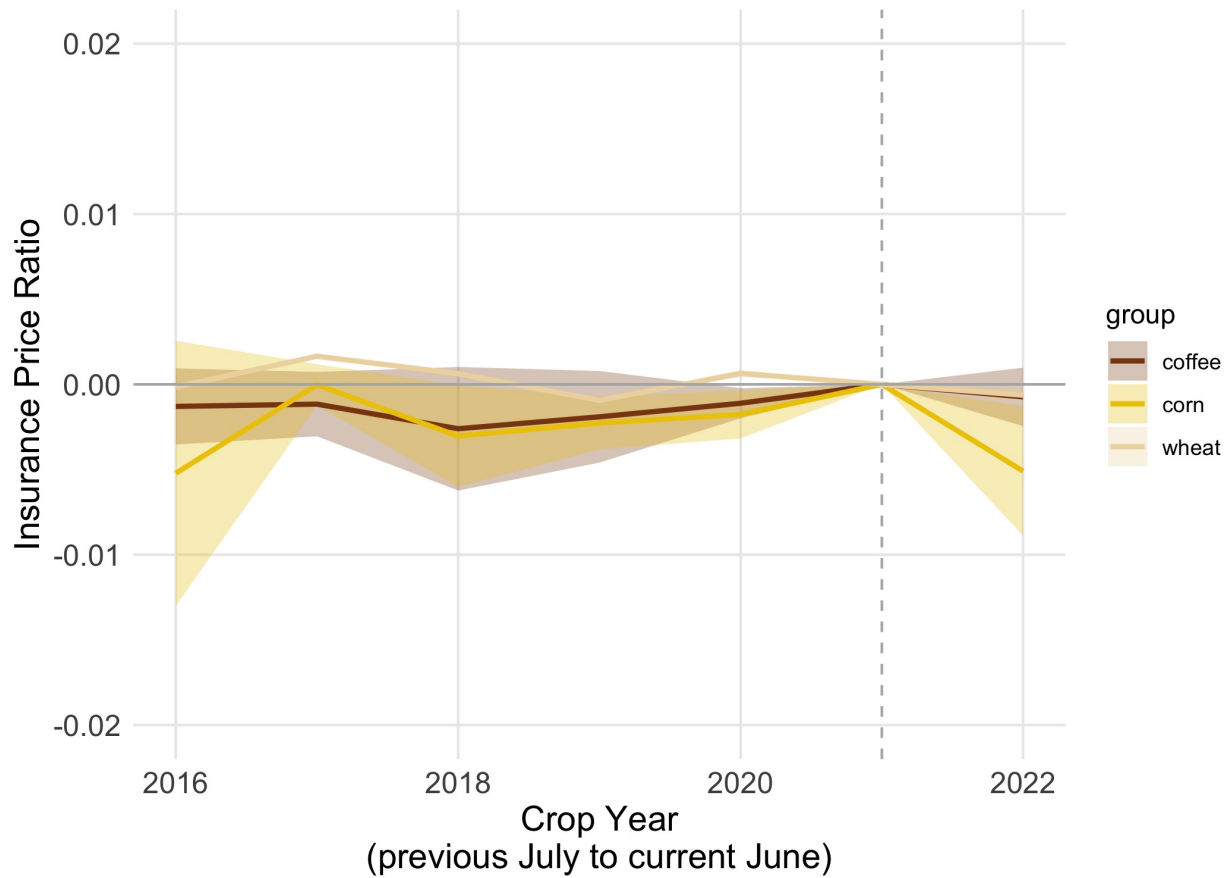
(a) No statistically significant effects on uninsured farmers' default rate.



(b) No statistically significant effects on insured farmers' default rate.

Notes: These BCB credit-registry regressions compare more and less damaged farmers relative to 2021Q2. The outcome is debt more than 90 days past due divided by total outstanding debt. Panel (a) reports uninsured farmers and panel (b) insured farmers. Specifications include farmer–lender and lender–municipality–quarter fixed effects. Bars report 95% confidence intervals based on standard errors clustered by farmer.

Figure A7: Insurance Premium of Frost-Affected Farmers to Unaffected



Notes: These SICOR regressions correspond to equation (31) for coffee, corn, and wheat. The outcome is the farmer-paid premium divided by insured value. Coefficients compare observed policies on more and less damaged plots relative to the 2021 growing season. Specifications include farmer and municipality–crop–year fixed effects. Bars report 95% confidence intervals based on standard errors clustered by farmer.

B Data Appendix

We combine all electronic payments at the Central Bank of Brazil: boletos, bank transfers, and instant payments. Because boletos can be settled via cash, and it is standard to pay by boleto when issuing a nota fiscal for tax purposes, we believe that we cover almost all of the farmers' input payments. We classify the following CNAE codes as upstream of coffee farming, whose code is 0134-2/00, for the payment regressions that we interpret as input purchases proxying for investment:

- **0141-5/01:** Production of certified seeds, except forage for pasture.
- **0142-3/00:** Production of certified coffee seedlings, including genetically modified

ones.

- **0161-0/02:** Pruning services in coffee plantations.
- **0161-0/03:** Contracted land preparation, cultivation, and harvesting services.
- **2831-3/00:** Manufacture of machinery and equipment for agriculture and livestock, parts and accessories.
- **2832-1/00:** Manufacture of machine tools, parts, and accessories.
- **3314-7/11:** Maintenance and repair of machinery and equipment for agriculture and livestock.
- **3314-7/12:** Maintenance and repair of machine tools.
- **4530-7/05:** Retail trade of tires and inner tubes.
- **4661-3/00:** Wholesale trade of machinery, devices, and equipment for agricultural use; parts and pieces.
- **7731-4/00:** Rental of agricultural machinery and equipment without operator.
- **7739-0/99:** Rental of other machinery and equipment not previously specified, without operator.
- **7490-1/03:** Intermediation and agency activities of services and businesses in general, except real estate.
- **0159-8/99:** Production of other plant-based products not previously specified.
- **0161-0/99:** Support activities for agriculture not previously specified.
- **4611-7/00:** Commercial representatives and trade agents of agricultural raw materials and live animals.

C Model Appendix

C.1 Sequential implementation of the unified collateral constraint

This appendix derives the debt–insurance representation of the unified collateral constraint, presents the household optimality conditions, and describes the numerical and estimation procedures for the model in Section 2.

For a farmer of type α , define harvest revenue and the appraised land value by

$$x_\alpha(S', z'; k) \equiv p(S') A_\alpha \theta(S', z')^\alpha k^\alpha, \quad (36)$$

$$q(S', z'; k) \equiv \bar{L} + \psi \kappa_G p(S') \theta(S', z') k. \quad (37)$$

We assume that, after default, the lender can recover fraction λ_K of surviving productive capital, fraction λ of harvest revenue, and a claim equal to fraction χ of appraised land value. The maximum enforceable state-contingent claim is therefore

$$\mathcal{B}_\alpha(S', z'; k) = \lambda_K \theta(S', z') k + \lambda x_\alpha(S', z'; k) + \chi q(S', z'; k). \quad (38)$$

The defaulting farmer retains the remaining value and can return to credit markets in subsequent periods. Because land is a fixed endowment in the current model, its appraised value affects the borrowing constraint but not the flow resource constraint. Thus χq is a reduced-form recovery claim rather than an explicit law of motion for land ownership.

Let $d \geq 0$ denote one-period non-state-contingent debt and $h(S', z') \geq 0$ a state-contingent insurance payoff. Define net repayment as

$$b(S', z') \equiv d - h(S', z'). \quad (39)$$

The debt–insurance constraint and the net-repayment constraint are equivalent:

$$\begin{aligned} d &\leq \mathcal{B}_\alpha(S', z'; k) + h(S', z'), & \forall (S', z'), \\ b(S', z') &\leq \mathcal{B}_\alpha(S', z'; k), & \forall (S', z'). \end{aligned} \quad (40)$$

Given any bounded schedule b , set

$$d = \left[\max_{S', z'} b(S', z') \right]^+, \quad h(S', z') = d - b(S', z'). \quad (41)$$

Then $h \geq 0$, condition (40) is exactly $b \leq \mathcal{B}_a$, and current net funding is unchanged:

$$\frac{d}{R} - \mathbb{E} \left[\frac{h(S', z')}{R} \right] = \mathbb{E} \left[\frac{b(S', z')}{R} \right]. \quad (42)$$

The reverse direction follows immediately from $b = d - h$. Thus the sequential debt and insurance contracts implement the net-repayment schedule in Proposition 1 for a given enforcement limit.

C.2 Optimality conditions

For a given perceived law of motion for prices, the farmer solves the following recursive problem. We omit the cross-sectional distribution from the notation.

$$V_a(W, S, z) = \max_{k, \{b(S', z')\}} \left\{ u(c) + \beta \sum_{S', z'} \pi(S', z' | S, z) V_a(W', S', z') \right\} \quad (43)$$

$$\text{s.t.} \quad W' = x_a(S', z'; k) + \theta(S', z')k - b(S', z'), \quad (44)$$

$$c = W + \sum_{S', z'} \pi(S', z' | S, z) \frac{b(S', z')}{R} - k, \quad (45)$$

$$b(S', z') \leq \mathcal{B}_a(S', z'; k) \quad \forall (S', z'). \quad (46)$$

Let $\xi(S', z') \geq 0$ be the multiplier on (46). Let $\mathbb{E}'$ denote expectation over (S', z') conditional on (S, z) , let $\pi'_{S', z'} \equiv \pi(S', z' | S, z)$, and write

$$\begin{aligned} x_{a,k}(S', z'; k) &= \alpha p(S') A_a \theta(S', z')^\alpha k^{\alpha-1}, \\ \mathcal{B}_{a,k}(S', z'; k) &= \lambda_K \theta(S', z') + \lambda x_{a,k}(S', z'; k) + \chi \psi \kappa_G p(S') \theta(S', z'). \end{aligned}$$

An interior solution satisfies

$$\begin{aligned} u'(c) &= \beta \mathbb{E}' \left[V_{a,W}(W', S', z') [x_{a,k}(S', z'; k) + \theta(S', z')] \right] \\ &\quad + \sum_{S', z'} \xi(S', z') \mathcal{B}_{a,k}(S', z'; k), \end{aligned} \quad (47)$$

$$\beta V_{a,W}(W', S', z') + \frac{\xi(S', z')}{\pi'_{S', z'}} = \frac{u'(c)}{R}, \quad (48)$$

$$V_{a,W}(W, S, z) = u'(c), \quad (49)$$

$$\xi(S', z') [\mathcal{B}_a(S', z'; k) - b(S', z')] = 0. \quad (50)$$

The bond and complementary-slackness conditions hold for every reachable (S', z') with $\pi'_{S', z'} > 0$; zero-probability states are excluded from the choice vector. The expectation in (47) integrates over continuation states. Individual farmers take $p(S')$ as given; the outer equilibrium loop updates it from aggregate output.

When the constraint binds, substituting (38) into (44) gives

$$W_a^B(S', z') = (1 - \lambda)p(S')A_a\theta(S', z')^\alpha k^\alpha + [1 - \lambda_k - \chi\psi\kappa_G p(S')] \theta(S', z')k - \chi\bar{L}. \quad (51)$$

In Regime II-a, $\psi = 0$, so land shifts the level of the limit but not the marginal collateral value of capital. In Regime II-b, surviving capital also creates pledgeable grove value. We impose the normal-state normalization in (7) when comparing the two regimes. Restriction (6) ensures that binding-state wealth remains weakly increasing in surviving capital.

C.3 Quantitative Scope and Validation

The equations above characterize the recursive equilibrium and isolate the budget and collateral-price channels. The quantitative results in the paper come from two solved fixed-price environments: the four-node CPR-and-land regime comparison in Section 2.5 and the capital-constraint benchmark below. In both exercises, the price path is an input common to all farmers and collateral regimes. This design isolates limited pledgeability from the equilibrium response of the coffee price.

The numerical implementation enumerates candidate binding sets and retains only solutions satisfying feasibility and complementary slackness in every reachable state. It checks positive consumption, the current and continuation budget identities, the no-money-pump restriction in (6), and nesting of the capital, CPR, and land constraints. Simulated event studies use fixed farmer draws and shock assignments across regimes, so differences in the reported paths arise from the collateral structure.

The parameter values in Tables 1, 2, and 6 are calibrated inputs rather than structural estimates. We evaluate the mechanism by applying the same outcome construction and event-study transformation to the empirical and simulated panels. The 20% pre-shock insurance share determines the mass assigned to the higher-coverage group; it does not identify a preference or enforcement parameter.

C.4 Solved Capital-Constraint Partial-Equilibrium Benchmark

The paper reports two distinct solved fixed-price exercises. Section 2.5 uses the four-node CPR household problem and separates farmers by initial model coverage. This subsection reports the two-state, all-uninsured capital-constraint benchmark. We report its optimal-

contract derivation, numerical solution, and benchmark parameterization because it provides additional quantitative evidence for the mechanism. To recover its Bellman equation from the unified environment, set $\lambda = \chi = 0$, set λ_K equal to the capital recovery parameter, and take the price process as given. The notation is linked by

$$\tilde{p}(s') = p(s')A_a \left[\theta^{\text{GE}}(s') \right]^\alpha, \quad \theta^{\text{GE}}(s') = (1 - \delta)\theta^{\text{PE}}(s'). \quad (52)$$

The recursive CPR-and-land environment instead sets $\lambda_K = 0$ and closes the coffee price through aggregate output and inverse demand.

C.4.1 Optimal Contract

We can write the optimal contract problem as a problem of choosing a sequence $\{c(s^t), t(s^t), k^*(s^t)\}_{t \geq \tau}$ to maximize,

$$\mathbb{E}_\tau \left[\sum_{t=\tau}^{\infty} \beta^{(t-\tau)} u(c_t) \right] \quad (53)$$

subject to the current and future period budget constraints:

$$W(s^\tau) \geq c(s^\tau) + t(s^\tau) + k^*(s^\tau), \quad (54)$$

$$\tilde{p}(s^t)k^*(s^{t-1})^\alpha + (1 - \delta)\theta(s^t)k^*(s^{t-1}) \geq c(s^t) + t(s^t) + k^*(s^t), \quad \forall t > \tau, \quad (55)$$

the lender participation constraint:

$$\mathbb{E}_\tau \left[\sum_{t=\tau}^{\infty} R^{-(t-\tau)} t_t \right] \geq 0 \quad (56)$$

where the transfer plan is restricted so that its discounted value is well-defined. For the debt representation below, we impose the transversality condition

$$\lim_{T \rightarrow \infty} R^{-(T+1-\tau)} \mathbb{E}_\tau [b(s^{T+1})] = 0. \quad (57)$$

and the limited commitment constraint:

$$\mathbb{E}_{\tau'} \left[\sum_{t=\tau'}^{\infty} \beta^{(t-\tau')} u(c_t) \right] \geq \mathbb{E}_{\tau'} \left[\sum_{t=\tau'}^{\infty} \beta^{(t-\tau')} u(\hat{c}_t) \right], \quad \forall \tau' \geq \tau, \forall \{\hat{c}(s^t)\}, \quad (58)$$

where $\left\{ \hat{c}(s^t) \right\}_{t=\tau'}^{\infty}$ is a solution to the optimal contract when the state variable W is given by

$$\hat{W}(s^{\tau'}) = \tilde{p}(s^{\tau'})k^*(s^{\tau'-1})^{\alpha} + (1-\lambda)(1-\delta)\theta(s^{\tau'})k^*(s^{\tau'-1}).$$

Endogenous debt limit framework We can write the farmer problem as:

of choosing a sequence $\{c(s^t), b(s^t), k^*(s^t)\}_{t \geq \tau}$ to maximize,

$$U(\{c(s^t)\} | s^{\tau}) := E_{\tau} \left[\sum_{t=\tau}^{\infty} \beta^{(t-\tau)} u(c_t) \right] \quad (59)$$

subject to the sequential budget constraint

$$\begin{aligned} & \tilde{p}(s^t)k^*(s^{t-1})^{\alpha} + (1-\delta)\theta(s^t)k^*(s^{t-1}) + \sum_{s^{t+1}} \frac{1}{R} \pi(s^{t+1}|s^t)b(s^{t+1}) \geq \\ & c(s^t) + k^*(s^t) + b(s^t), \quad \forall t \geq \tau, \end{aligned} \quad (60)$$

and the debt limit is given by a process $\{\tilde{D}^i(s^t)\}_{t > \tau}$:

$$\lambda(1-\delta)\theta(s^t)k^*(s^{t-1}) - b(s^t) \geq -\tilde{D}^i(s^t), \quad \forall t > \tau. \quad (61)$$

On the following, we define the continuation utility, conditional on repaying debt equal to x at state s^{τ} and having debt limit given by $\{\tilde{D}^i(s^t)\}_{t > \tau}$ as $\tilde{V}^i(\{\tilde{D}^i(s^t)\}_{t > \tau}, x | s^{\tau})$. We say that a sequence of debt limit is *not too-tight* if for all s^t ,

$$\tilde{V}^i(\{\tilde{D}^i(s^t)\}_{t > \tau}, x | s^{\tau}) = \tilde{V}^i(\{\tilde{D}^i(s^t)\}_{t > \tau}, 0 | s^{\tau})$$

Following the arguments from Section 3.2 of Martins-Da-Rocha et al. (2022), we can show that a sequence of debt limits is not too tight if and only if $\tilde{D}^i(s^t) = 0$ for all s^t .

Lemma 1. (*Endogenously incomplete markets*) *A consumption allocation is the outcome of the optimal contract if and only if, it is the outcome of the endogenous debt limit framework.*

Proof. ($\Rightarrow$) Fix a sequence $\{c(s^t), t(s^t), k^*(s^t)\}_{t \geq \tau}$ that solves the farmer's optimal contract problem. We first show that if a sequence $\{t(s^t)\}_{t \geq \tau}$ satisfies (56) and (58), then

$$\lambda(1-\delta)\theta(s^{\tau'})k^*(s^{\tau'-1}) \geq E_{\tau'} \left[\sum_{t=\tau'}^{\infty} \frac{t_t}{R^{t-\tau'}} \right], \quad \text{for all } \tau' \geq \tau. \quad (62)$$

Assume by way of contradiction that there exists $\tau', s^{\tau'}$ such that

$$\lambda(1 - \delta)\theta(s^{\tau'})k^*(s^{\tau'-1}) < \mathbb{E}_{\tau'} \left[\sum_{t=\tau'}^{\infty} \frac{t_t}{R^{t-\tau'}} \right].$$

Consider the following deviation. The farmer defaults at τ' and enters a contract with the following allocation schedule:

$$\hat{t}(s^t) = \begin{cases} t(s^t) & \text{if } t > \tau' \\ -\mathbb{E}_{\tau'} \left[\sum_{t=\tau'+1}^{\infty} \frac{t_t}{R^{t-\tau'}} \right] & \text{if } t = \tau' \end{cases}$$

$$\hat{k}^*(s^t) = k^*(s^t), \quad \text{for all } t \geq \tau'.$$

By construction, $\hat{c}(s^t) = c(s^t)$ for all $t > \tau'$. We choose $\hat{c}_{\tau'}$ so that the budget constraint holds at τ' . Moreover, the definition of $\hat{t}$ ensures that (56) holds. The change in consumption at the default date is

$$\begin{aligned} \hat{c}_{\tau'} - c_{\tau'} &= t(s^{\tau'}) - \hat{t}(s^{\tau'}) - \lambda(1 - \delta)\theta(s^{\tau'})k^*(s^{\tau'-1}) \\ &= \mathbb{E}_{\tau'} \left[\sum_{t=\tau'}^{\infty} \frac{t_t}{R^{t-\tau'}} \right] - \lambda(1 - \delta)\theta(s^{\tau'})k^*(s^{\tau'-1}) > 0, \end{aligned}$$

which contradicts optimality. The remainder of the proof is analogous to Rampini and Viswanathan (2013). Define

$$b(s^{\tau'}) := \mathbb{E}_{\tau'} \left[\sum_{t=\tau'}^{\infty} R^{-(t-\tau')} t_t \right] \leq \lambda(1 - \delta)\theta(s^{\tau'})k^*(s^{\tau'-1}), \quad \forall \tau' > \tau.$$

Then $t_{\tau'} = b(s^{\tau'}) - \sum_{s^{\tau'+1}} \frac{1}{R} \pi(s^{\tau'+1}|s^{\tau'}) b(s^{\tau'+1})$ for all $\tau' \geq \tau$, with $b(\tau) = 0$. Then the budget constraints (60) become equivalent to the budget constraints (54). Moreover, (61) holds, so the sequence of consumption is feasible in the economy with endogenous debt limit.

($\Leftarrow$) Fix a sequence $\{c(s^t), b(s^t), k^*(s^t)\}_{t \geq \tau}$ that solves the farmer's maximization problem under the endogenous debt limit framework. Define the repayment values, for all $t > \tau$:

$$t(s^t) := b(s^t) - \sum_{s^{t+1}} \frac{1}{R} \pi(s^{t+1}|s^t) b(s^{t+1}),$$

Consider the sequence $\{t(s^t)\}_{t \geq \tau}$. This sequence satisfies the budget constraint (54) by definition. Moreover, (56) is satisfied.¹³

$$\mathbb{E}_\tau \left[\sum_{t=\tau}^{\infty} R^{-(t-\tau)} t_t \right] = 0.$$

Now assume by way of contradiction that $\{t(s^t)\}_{t \geq \tau}$ does not satisfy (58). Then there exists $\tau' \geq \tau$ such that $\mathbb{U}(\{c(s^t)\} | s^{\tau'}) < \mathbb{U}(\{\hat{c}(s^t)\} | s^{\tau'})$, where $\{\hat{c}(s^t)\}$ solves the optimal contract problem with initial net worth $\tilde{p}(s^{\tau'})k^*(s^{\tau'-1})^\alpha + (1-\lambda)(1-\delta)\theta(s^{\tau'})k^*(s^{\tau'-1})$. Let the associated transfers be $\hat{t}(s^t)$. By the previous part of the proof,

$$\hat{b}(s^t) := \mathbb{E}_t \left[\sum_{k=t}^{\infty} R^{-(k-t)} \hat{t}_k \right], \quad \forall t > \tau'$$

is feasible in the endogenous debt limit with $b(s^{\tau'}) = 0$. Now define the alternative consumption plan,

$$\tilde{c}(s^t) = \begin{cases} c(s^t) & \text{if } t < \tau', \\ \hat{c}(s^t) & \text{if } t \geq \tau'. \end{cases}$$

This consumption plan is feasible under the endogenous debt limit problem because at τ' we have $b(s^{\tau'}) \leq \lambda(1-\delta)\theta(s^{\tau'})k^*(s^{\tau'-1})$. It follows that $\mathbb{U}(\{c(s^t)\} | s^\tau) < \mathbb{U}(\{\tilde{c}(s^t)\} | s^\tau)$, contradicting optimality. $\square$

Corollary 1. (*Alternative characterization of endogenously incomplete markets*) *A consumption allocation is the outcome of the optimal contract if and only if the allocation is the outcome of an economy where farmers only have access to a sequence of one-period state contingent savings contracts $\{h(s^t)\}_{t \geq \tau}$ and one period (not state contingent) debt contracts $\{d(s^t)\}_{t \geq \tau}$, satisfying :*

$$\lambda(1-\delta)\theta(s^{\tau'})k^*(s^{\tau'-1}) + h(s^{\tau'}) \geq d(s^{\tau'-1}), \quad \forall \tau' > \tau \quad (63)$$

$$d(s^t) \geq 0, \quad h(s^t) \geq 0, \quad \text{for all } t > \tau$$

¹³For every finite T , iterated expectations give

$$\mathbb{E}_\tau \left[\sum_{t=\tau}^T R^{-(t-\tau)} t_t \right] = b(s^\tau) - R^{-(T+1-\tau)} \mathbb{E}_\tau [b(s^{T+1})].$$

We set $b(s^\tau) = 0$. Taking $T \rightarrow \infty$ under condition (57) gives a lender value of zero.

Proof. We obtain the result by defining, for every $\tau' > \tau$ and every history $s^{\tau'}$,

$$d(s^{\tau'-1}) := \left[\max_{s^{\tau'}} b(s^{\tau'}) \right]^+$$

$$h(s^{\tau'}) := d(s^{\tau'-1}) - b(s^{\tau'}),$$

and appeal to Lemma 1. □

C.4.2 General Solution Technique

Below is an algorithm to solve the problem

$$\begin{aligned} V(W, s) &= \max_{c, k, \{W(s')\}, \{b(s')\}} u(c) + \beta \mathbb{E}[V(W(s'), s') \mid s] \\ \text{s.t. } W + \sum_{s'} \pi(s' \mid s) \frac{b(s')}{R} &\geq c + k, \\ \tilde{p}(s')k^\alpha + (1 - \delta)\theta(s')k &\geq W(s') + b(s'), \quad \text{for all } s', \\ \lambda(1 - \delta)\theta(s')k &\geq b(s'), \quad \text{for all } s'. \end{aligned}$$

Note that we can rewrite this equation as

$$\begin{aligned} V(W, s) &= \max_{k, \{b(s')\}} u \left(W + \sum_{s'} \pi(s' \mid s) \frac{b(s')}{R} - k \right) \\ &\quad + \beta \sum_{s'} \pi(s' \mid s) V(\tilde{p}(s')k^\alpha + (1 - \delta)\theta(s')k - b(s'), s') \\ \text{s.t. } \lambda(1 - \delta)\theta(s')k &\geq b(s'), \quad \text{for all } s'. \end{aligned}$$

For compactness, define

$$q(s') \equiv (1 - \delta)\theta(s'), \quad Y(s'; k) \equiv \tilde{p}(s')k^\alpha + q(s')k.$$

The algorithm is as follows:

- **Inputs:** Parameters, a grid $\mathcal{W} = (W_l, \dots, W_h)$ for next-period wealth, an initial guess for $V_W(W, s)$, and an interpolation rule outside the grid.
- In the unconstrained case, capital satisfies

$$k^*(s) = \left(\frac{\alpha \sum_{s'} \pi(s' \mid s) \tilde{p}(s')}{R - \sum_{s'} \pi(s' \mid s) q(s')} \right)^{\frac{1}{1-\alpha}}.$$

When no continuation-state constraint binds, the marginal-utility ratio is degenerate at $k^*(s)$. We recover $u'(c)$ from $u'(c) = \beta R V_W(W(s_j), s_j)$ at the grid anchor and then recover wealth in the remaining states. For a binding anchor state s_j and a grid value $W(s_j) = w_g$, recover k from

$$w_g = \tilde{p}(s_j)k^\alpha + (1 - \lambda)q(s_j)k.$$

- For each current state s , grid value w_g , and candidate set $\mathcal{B}$ of binding continuation-state constraints:

1. For each $s_j \in \mathcal{B}$, impose

$$W(s_j) = Y(s_j; k) - \lambda q(s_j)k = \tilde{p}(s_j)k^\alpha + (1 - \lambda)q(s_j)k.$$

2. Compute marginal utility from

$$u'(c) = \frac{\beta R \sum_{s_j \in \mathcal{B}} \pi(s_j | s) V_W(W(s_j), s_j) [\alpha \tilde{p}(s_j)k^{\alpha-1} + (1 - \lambda)q(s_j)]}{R - \sum_{s_j \in \mathcal{B}} \pi(s_j | s) \lambda q(s_j) - \sum_{s_j \notin \mathcal{B}} \pi(s_j | s) [\alpha \tilde{p}(s_j)k^{\alpha-1} + q(s_j)]}.$$

Recover c by inverting u' .

3. For each $s_j \notin \mathcal{B}$, find $W(s_j)$ such that

$$\beta R V_W(W(s_j), s_j) = u'(c).$$

4. Recover repayment in every state from

$$b(s_j) = Y(s_j; k) - W(s_j).$$

Retain the candidate only if $b(s_j) = \lambda q(s_j)k$ for binding states and $b(s_j) < \lambda q(s_j)k$ for slack states.

5. Recover current wealth from

$$W = c + k - \sum_j \frac{\pi(s_j | s) b(s_j)}{R},$$

and update $V_W^{\text{next}}(W, s) = u'(c)$.

- Update V_W and repeat until $\|V_W - V_W^{\text{next}}\|_{\infty, \mathcal{W}}$ is below tolerance.

C.4.3 Numerical Simulation

We consider a continuum of farmers with unit mass. Each farmer receives an idiosyncratic shock to capital survival, which takes values $\theta = 0.3$ in the bad state and $\theta = 0.95$ in the good state. The benchmark features two aggregate states. In the good aggregate state, 20% of farmers receive the bad idiosyncratic shock. In the bad aggregate state, 50% of farmers receive the bad idiosyncratic shock.

We use $\alpha = 0.3$, $\lambda = 0.7$, $\beta = 0.9$, $R = 1.05$, and CRRA utility with $\sigma = 2$. Within this solved benchmark, λ is the capital recovery parameter denoted by λ_K in the unified constraint. The transition matrix is set so that the unconditional distribution over aggregate states is

$$P_S = [0.8, 0.2]. \quad (64)$$

Table 6: Parameters in the Solved Partial-Equilibrium Benchmark

Parameter	Value	Interpretation
α	0.30	Capital elasticity
λ_K	0.70	Recoverable share of productive capital
β	0.90	Discount factor
R	1.05	Gross funding rate
σ	2.00	CRRA coefficient
θ_f	0.30	Capital survival in the bad state
θ_n	0.95	Capital survival in the good state

The simulation design uses 11,000 periods with 10,000 farmers and discards the first 1,000 periods. Define $\text{Shock}_{it} = 1$ if the aggregate state is bad in period t and farmer i receives the idiosyncratic shock. The simulated-panel regression is

$$y_{it} = \sum_{\tau=-G}^M \beta_{\tau}^{\text{NI}} \text{Shock}_{i\tau} + \alpha_i + \alpha_t + \varepsilon_{it}, \quad (65)$$

where all farmers in this simulation are uninsured. Section 2.5 reports the separate solved fixed-price CPR exercise with distinct initial model-coverage groups. Together, the two exercises show how limited pledgeability affects both the short-run response and the persistence of rebuilding.

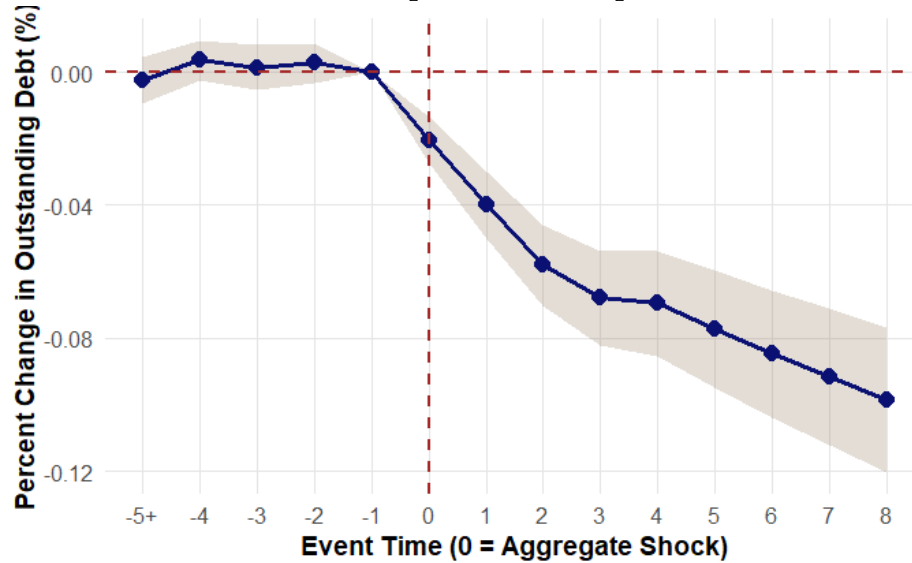
C.4.4 Additional Numerical Results

Section 2.5 reports the insured and uninsured responses from the fixed-price CPR exercise used to motivate the empirical specifications. For completeness, Figures C1 and C2

report the longer-horizon results from the two-state capital-constraint benchmark. In this simulation, all farmers in the event-study regression are uninsured.

We compute the cumulative responses of farmers' debt load to the shock using the model-implied event-study coefficients. Figure C1 shows a persistent decline in borrowing following the shock. The frost reduces the capital that the farmer can pledge, and the borrowing constraint remains tight during the recovery.

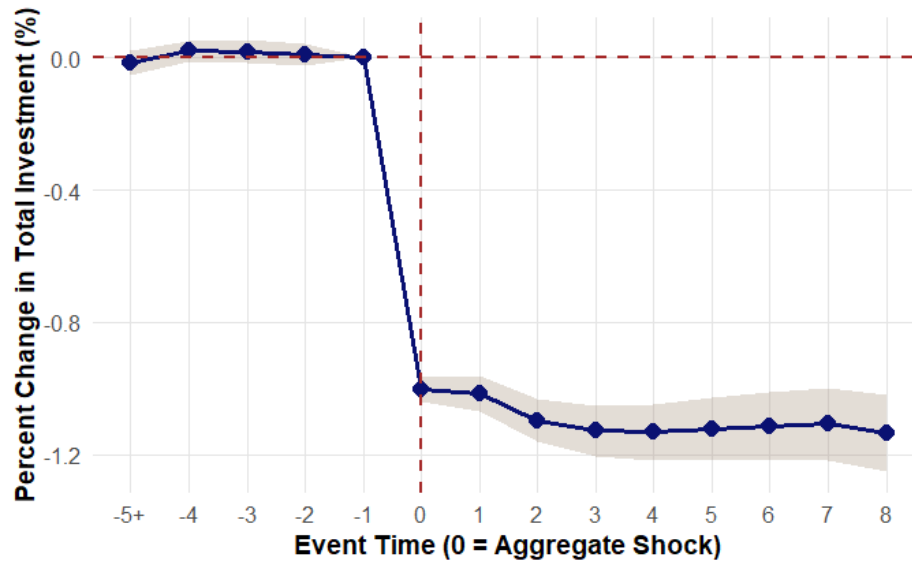
Figure C1: Cumulative Model-Implied Debt Response to an Adverse Shock



Notes: Based on model-simulated data, we estimate the cumulative effect of an adverse shock on debt using a fixed-effects event study. The estimated coefficients capture the dynamic adjustment in debt relative to the quarter of the shock. All farmers in this simulation are uninsured.

Figure C2 shows a prolonged decline in investment following the shock. This pattern is the opposite of what one would expect under complete markets, because the farmers affected by the shock have a higher marginal product of capital and should invest more to replenish their capital stock. Under limited pledgeability, the loss of capital tightens the constraint and prevents this adjustment.

Figure C2: Cumulative Model-Implied Investment Response to an Adverse Shock



Notes: The plot shows the dynamic response of investment following a negative shock, based on the same simulated economy used for the debt results. The specification mirrors that of the debt response, with farmer and time fixed effects.